

3

Stars and the Universe

In this chapter, we will learn about

- the method of parallax
- the apparent brightness, luminosity, apparent magnitude and absolute magnitude of stars
- the relationship between the energy radiated by a star, its colour and its size
- the Doppler effect of light as an indication of the motion of celestial bodies

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3.1 Measuring distances of stars

The flower and the man in Figure 3.1a appear very big.



Fig 3.1a *Objects in the photos appear larger than they actually are.*

This can be explained using Figure 3.1b.

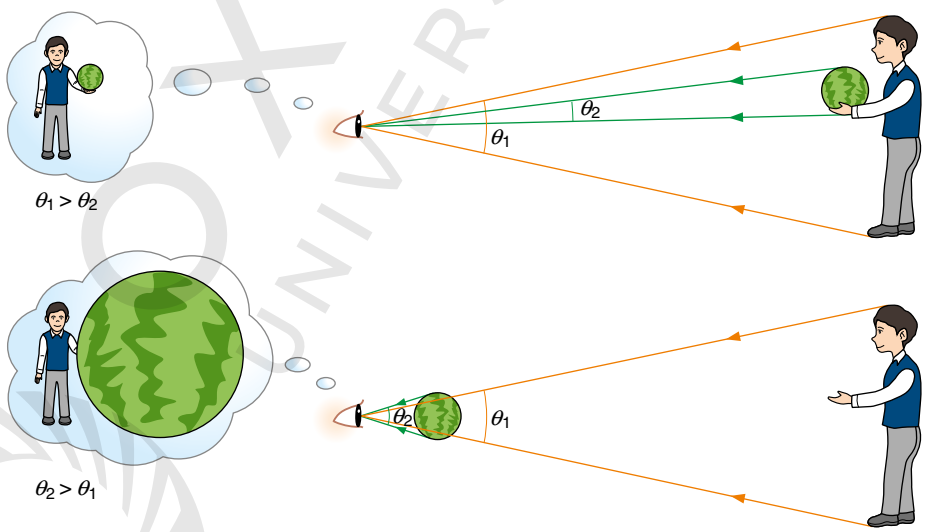


Fig 3.1b *The apparent size of the boy and the watermelon depends on the angle subtended at the eye of the observer.*

When an object occupies a larger angle of view, it appears bigger. This principle is applied in astronomy. Since the apparent sizes and separations of objects in the sky cannot be measured using a ruler, they are described as angles instead.

In astronomy, angles can also be used to deduce the distances of stars from us. Before learning how such distances are calculated, let us first learn some more about angles.

1 Angular measurement

a Common units of angle

The units of angles include the degree (°) and there are 360° in a circle. Smaller angles are measured in **minutes of arc** (**arcmin** or ′) or **seconds of arc** (**arcsec** or ″).

Skill *Mathematics*

Conversion between minute/second and decimal ▸ p.157

$$1' = 60'' \quad \text{and} \quad 1° = 60' = 3600''$$

Radian (rad) is another unit for measuring angles (Fig 3.1c). There are 2π rad in a full circle.

$$\therefore 1 \text{ rad} = \frac{360°}{2\pi} = \frac{180°}{\pi} = 57.3°$$

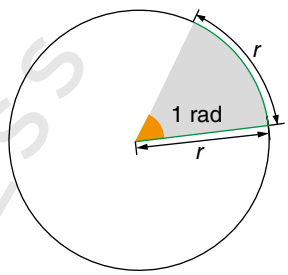


Fig 3.1c *Definition of 1 rad.*

b Angular size and angular distance

Angular measurements do not represent actual distances or sizes of the celestial bodies.

The **angular size** (or **angular diameter**) describes the apparent size of a **single** object in the sky; while the **angular distance** describes the apparent distance between **two** objects in the sky (Fig 3.1d).

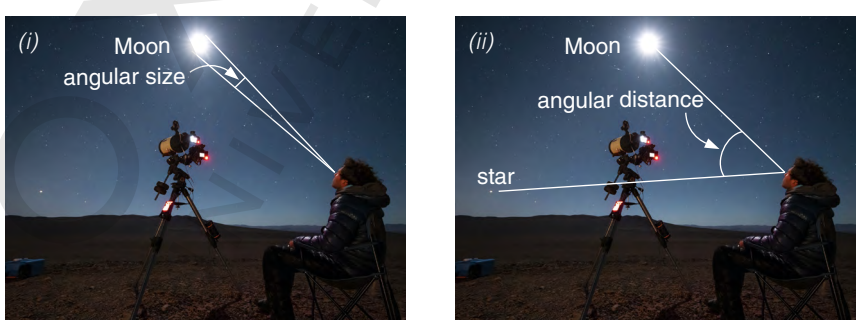
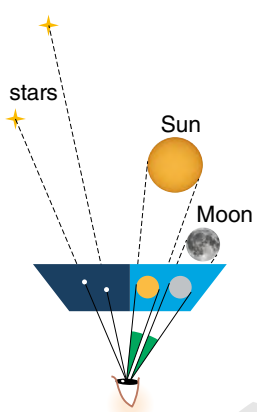


Fig 3.1d *The meaning of (i) angular size and (ii) angular distance.*



The Sun and the Moon are different in actual sizes but have roughly the same angular sizes in the sky. This is the reason why both total solar eclipse and annular solar eclipse are possible.

We can estimate angular sizes or distances of objects in the sky by holding our hand at arm's length (Fig 3.1e).

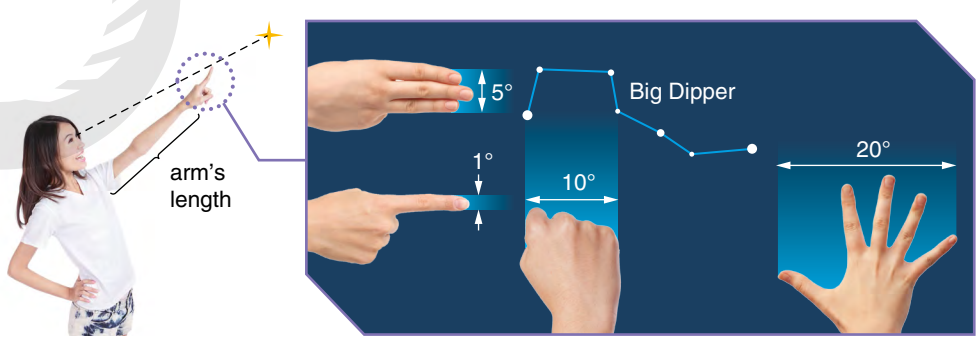


Fig 3.1e *Estimating angular sizes/distances using your hand held at arm's length.*

minute of arc (arcmin or ′) 角分　　second of arc (arcsec or ″) 角秒　　angular size 角大小
angular diameter 角直徑　　angular distance 角距離

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 *True or False:*

(a) 15′ = 0.25° (T/F)

(b) 1.55° = 1°30′30″ (T/F)

(c) Two objects are of the same size if the ratios of their angular size to their distance from us are equal. (T/F)

2 Complete the following conversion.

	Angle in degrees	Angle in radians
(a)		0.027
(b)	9.40	
(c)	45.6	
(d)		1.23

DSE goal

Determine the distance of a celestial body using the method of parallax.

DSE exam

13(2)Q1.4, 17(2)Q1.4, 18(2)Q1.5, 19(2)Q1.3, 20(2)Q1.7, 23(2)Q.1(d)

Video

→ This video demonstrates the Hands-on activity.

2 Parallax method

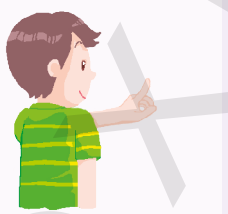
Hands-on activity

Seeing with one eye

→ Worksheet is available on OUP web.

What to do

1 Hold your finger out at arm's length.



2 Close your left eye. Look at the finger with your right eye.



3 Now open your left eye and close the other. Look at the finger again.

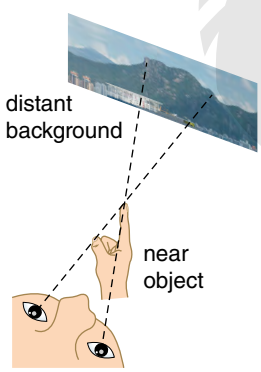


What you find

Does the position of your finger change against background objects?

Simulation

→ This simulation demonstrates parallax as we observe objects from different positions. It is available on OUP web in PC version (.exe) only.



The phenomenon you experienced in the activity is called **parallax**. It is an apparent change in the position of a near object against the distant background due to a change in viewpoints. Figure 3.1f shows an example on what you should see under the parallax effect. The image of your finger should shift relative to the background.

(i) The view of right eye only. *(ii) The view of left eye only.*



Fig 3.1f *The position of your finger appears to change against background objects under the parallax effect.*

parallax 視差

The method of parallax is one of the cleanest way to measure astronomical distances, as it makes very few assumptions.

Astronomers can use the parallax method to measure distances to **nearby** stars, simply by taking advantage of the different positions of the Earth during its orbital motion around the Sun.

The observed shift is the largest when the star is observed from two points on opposite sides of the Earth's orbit (Fig 3.1g). To measure the distance to a nearby star, we take photos of the star on many nights throughout the year, and find its maximum angular shift against the background stars.

The background stars are very far away from the Earth that their parallaxes are too small to be noticeable. Therefore, they can be regarded as fixed.

The change in position of the nearby star is only apparent. We assume the actual motion of the star is negligible.

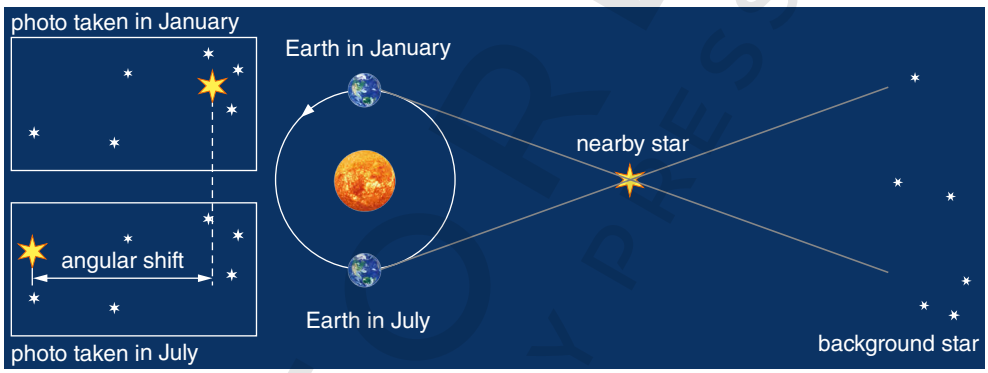


Fig 3.1g *The parallax effect is maximized when observing a nearby star six months apart.*

It takes six months for the Earth to move from one side to the other, so the two photos of the maximum shift should have been taken six months apart. The **nearer** the star to us, the **larger** the maximum angular shift shown in the photos.

For a star with maximum angular shift of $2p''$ (Fig 3.1h), its **angle of stellar parallax**, or simply parallax, p'' is related to the distance d of the star by:

$$\frac{1\text{ AU}}{d} = \tan p'' \quad \text{..... (1)}$$

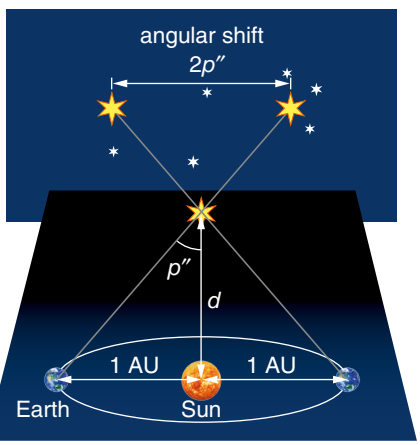


Fig 3.1h *The maximum angular shift is twice the angle of stellar parallax.*

Skill *Mathematics*

Small angle approximation ▸ p.158

To use the small angle approximation, we have to convert the angle from arcsecs to radians.

$$1\text{ rad} = \frac{180°}{\pi} = \frac{180 \times 3600''}{\pi}$$

$$= 206\,265''$$

$$\Rightarrow 1'' = \frac{1}{206\,265}\text{ rad}$$

Therefore, equation (1) becomes

$$\frac{1\text{ AU}}{d} = \tan p'' = \tan\left(\frac{p}{206\,265}\text{ rad}\right) \approx \frac{p}{206\,265}$$

$$\Rightarrow d = \frac{206\,265\text{ AU}}{p} \quad \text{..... (2)}$$

Know more

Other applications of parallax

Parallax gives us perception of depth. Apart from measuring stellar distances, it has other applications. Do you know what they are?

▸ p.155

angle of stellar parallax 恒星視差角

To make calculations easier, astronomers prefer to express the distance to a star in parsec (pc). One parsec is defined to be the distance to a star whose **par**allax is one arc**sec** (Fig 3.1i), i.e.

One parsec is also defined as the distance at which 1 AU subtends an angle of 1″. These two definitions are equivalent.

1 pc = 206 265 AU

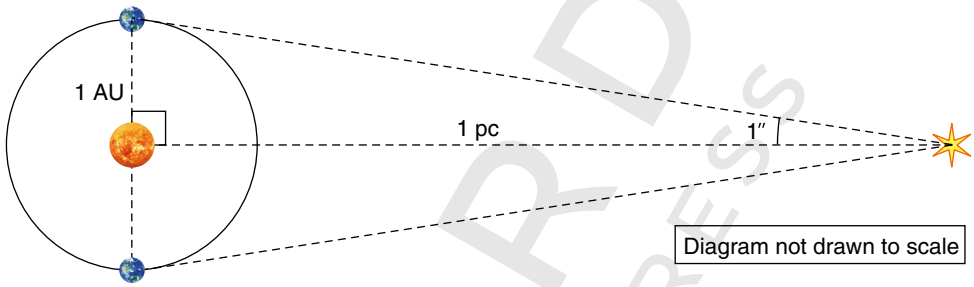


Fig 3.1i *Definition of one parsec.*

If p is measured in arcsecs and d is measured in parsecs, we can simplify equation (2) as:

$$d = \frac{1}{p}$$

(d: in parsec; p: in arcsec)

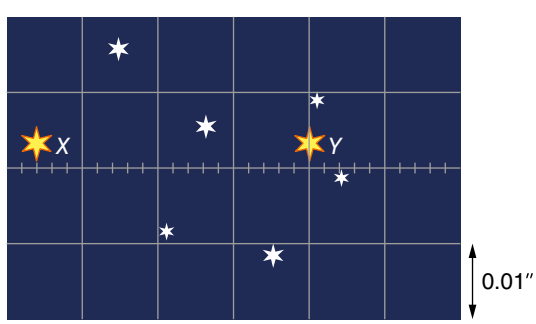
As a unit of distance, parsecs can be converted to AU, m and ly.

1 pc = 206 265 AU = 3.09×10^{16} m = 3.26 ly

Example 1 Finding the distance to a star using parallax

In astronomy, the symbol ⊙ denotes the Sun. Therefore, $R_\odot$ denotes the solar radius.

R Doradus, a star of radius $298R_\odot$, shows a maximum shift from position X to Y over one year relative to background stars as shown below.



(a) Estimate the distance to R Doradus.

(b) Estimate the angular size (in arcsec) of R Doradus in the sky.

Given: 1 pc = 3.09×10^{16} m, $R_\odot = 6.96 \times 10^8$ m

Solution

(a) From the figure, the maximum angular shift of R Doradus is 0.036″.

$$\text{Parallax } p = \frac{0.036''}{2} = 0.018''$$

$$\text{Distance to R Doradus} = \frac{1}{p} = \frac{1}{0.018} = 55.6 \text{ pc}$$

(b) $$\text{Angular size of R Doradus} = \frac{2 \times 298 \times 6.96 \times 10^8}{55.6 \times 3.09 \times 10^{16}}$$

$$= 2.414 \times 10^{-7} \text{ rad}$$

$$= 2.414 \times 10^{-7} \text{ rad} \times \frac{180 \times 3600''}{\pi}$$

$$= 0.0498''$$

R Doradus is regarded as the star (except the Sun) with the largest apparent size as viewed from the Earth. ▶

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) Parallax can be used to determine the distance of Jupiter from the Earth. (T/F)

(b) Background stars having negligible stellar parallax must be equidistant from the Earth. (T/F)

2 The longest distance below is

A 10 AU. B 1×10^{10} km.

C 5 ly. D 1 pc.

3 The information about some stars are shown below. Complete the table.

	Star	Parallax (in ″)	Distance (in pc)	Distance (in AU)
(a)	Sirius	0.379		
(b)	Altair			1.06×10^6
(c)	Vega		8	

Learn by practice 1

1 The positions of stars *A* and *B* in January and July observed from the Earth are as shown.

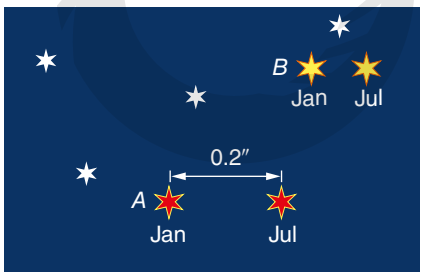


Fig a

(a) Which star, *A* or *B*, is further away from the Earth?

(b) The parallax of star *A* shown in Figure a is the maximum over one year. Find the distance (in pc) of star *A* from the Earth.

(c) The parallax of star *C* is twice that of star *B*. If the distance of star *B* from the Earth is *r*, what is the distance of star *C* in terms of *r*?

2 A star is 16.3 light years away from us. Estimate the maximum angular shift (in arcsecs) of this star in the sky throughout a year.

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☆ HKDSE 2017 Paper 2 Q1.4

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Movement of nearby stars in the sky

The pictures on the right show the same region of the sky. The positions of five stars, M, N, O, P and Q, are recorded on two different days.

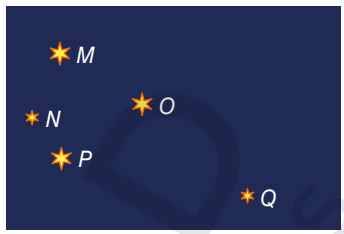
Which of the following statements about the stars must be correct?

A N and Q are further away from the Earth than M, O and P.

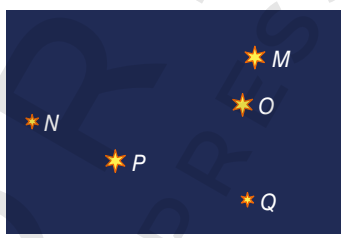
B N and Q are at the same distance away from the Earth.

C The parallax of M is the smallest.

D P is the closest to the Earth.



1 April



1 September

Solution

There is nearly no angular shift for N and Q. They can be treated as background stars. As their parallaxes are too small to be measured, we cannot compare the distances to these two stars.

∴ A is correct and B is not necessarily correct.

In the two pictures, the angular shift of M is the largest.

∴ C is incorrect.

By $d = \frac{1}{p}$, the smaller the parallax (or angular shift), the further away the star from us.

∴ D is incorrect.

Common mistake ⚠
Students may not know the nearer the star to us, the larger the angular shift is shown in the pictures.

▶ Revision 3 Q18 (p.148)

3 Limitations of the method of parallax

Proxima Centauri, the nearest star beyond the Sun, has a parallax of only 0.768″. Other stars have even smaller parallaxes.

From the definition of parsec, we can see that the greater the distance of the star, the smaller the parallax. Due to atmospheric distortion, parallax angles smaller than about 0.01″ are very difficult to be measured accurately by telescopes on the ground. Typically, distances up to 100 pc can be determined by the parallax method.

The measurable distance can be greatly extended if space telescopes are used. For example, the space telescope Gaia can measure up to about 9200 pc (30 000 ly).

The method of parallax has two main limitations.

1 It is time consuming as each measurement takes at least six months.

2 It cannot be used on distant stars as the parallaxes are too small to be measured.

Exercise 3.1

1 Which of the following units is used in expressing the distance to a star?

A Arcsec B Degree
C Parsec D Radian

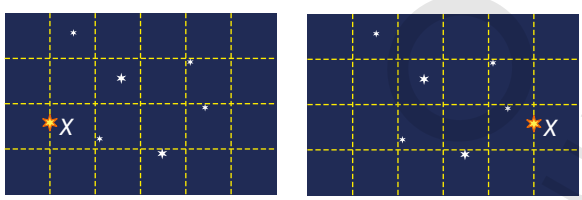
2 If the Earth's orbit around the Sun was smaller, how would the parallax of a star change?

A It would become smaller.
B It would be the same.
C It would be larger.
D The change cannot be determined since the distance to the star is unknown.

***3** Suppose a star is about 1 pc from the Sun. What would be the angular size of the Earth's orbit when it was observed from the star?

A 1° B 2′
C 1″ D 2″

4 Two pictures of the same region of the sky taken six months apart are shown below (Fig a). In this period, star X shows a maximum shift in position over the year. Each grid square corresponds to an angular scale of 0.1″. Find the distance to X in parsecs.



(i) 25 July. *(ii) 25 January.*

Fig a

***5** Stars X and Y have parallaxes of 0.1″ and 0.05″ respectively, as observed from the Earth.

(a) Which star is closer to the Earth?

(b) How much closer is the star in (a) than the other one?

(c) What would the parallax of star X be if it were measured on Neptune? The semi-major axis of Neptune is 30.1 AU and its orbit can be considered circular.

(d) The average orbital radius of Mars is 3 AU. How would having an observatory on Mars help measure the distance to stars more accurately?

***6** The Moon appears much larger than Jupiter in the sky (Fig b).



Fig b

(a) Which one has a larger angular size?

(b) Which one has a larger actual size?

(c) Explain why this phenomenon occurs.

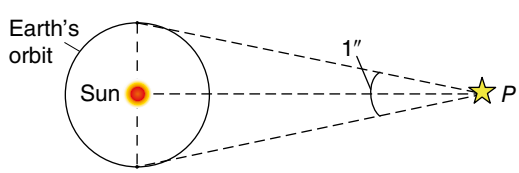
(d) The distance to the Moon is about 384 000 km. The angular separation between the Moon and Jupiter is about 2°. Annie wrongly estimates the actual separation between the Moon and Jupiter as shown (Fig c). State all the mistakes in her calculation.

Actual separation between the Moon and Jupiter
$= r\theta$
$= 3.84 \times 10^8 \times 2 \times \frac{\pi}{360}$
$= 6.70 \times 10^6$ m

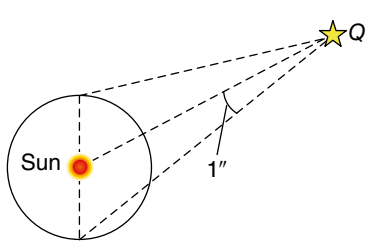
Fig c

***7** A student measures the distances to stars P, Q and R according to the figures below. He wrongly concludes that they are all 1 pc away. Explain what mistake(s) the student has made in each measurement. Assume the star in each case is on the plane of Earth's orbit.

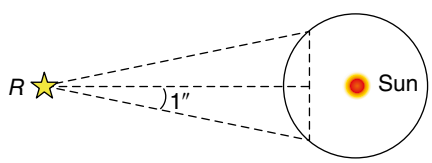
(a)



(b)



(c)



3.2 Brightness of stars

Street lamps in an array are often of the same power and hence have equal brightness, but the closer ones look brighter to us (Fig 3.2a). How about stars? Are brighter stars always closer to us?



Fig 3.2a An array of street lamps.

1 Luminosity and apparent brightness

Unlike the street lamps in an array, stars have different powers. The brightness of a star as seen between the star and the Earth is related to both the amount of energy emitted by the star and the distance of the star.

a Luminosity

In Book 5, we learned that the core of a star undergoes nuclear fusion and energy is released. The power radiated by a star is called its **luminosity**, denoted by L. It is the total energy given out by a star in the form of electromagnetic radiation per unit time in all directions.

$$\text{Luminosity } (L) = \frac{\text{total energy emitted}}{\text{time}}$$

Luminosity is measured in W or J s^{-1}. It is independent of the distance of the star from the Earth.

luminosity 光度

b Apparent brightness

The **apparent brightness**, denoted by b, measures the brightness of a star **as seen from the Earth**. It is the total energy coming from the star per second per square metre normal to the light ray as measured on Earth, i.e. the **power received per unit area**, and is measured in W m^{-2}.

Unit of $b = \frac{\text{J}}{\text{m}^2 \times \text{s}} = \frac{\text{J}}{\text{s}} \times \text{m}^{-2}$
$= \text{W m}^{-2}$

Consider a star of luminosity L radiating energy uniformly in all directions (Fig 3.2b).

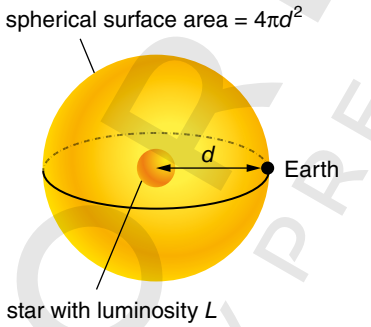


Fig 3.2b *Only a small portion of the energy given out by a star reaches the Earth.*

If the star is at a distance d away from the Earth,

$$\text{apparent brightness } b \text{ of the star} = \frac{\text{total energy}}{\text{time}} \times \frac{1}{\text{area}} = L \times \frac{1}{4\pi d^2}$$

Note that the total energy here refers to the light energy of the whole EM spectrum, not just the visible part.

apparent brightness (W m^{-2}) $b = \dfrac{L}{4\pi d^2}$ **luminosity** (W or J s^{-1}); **distance** between the star and the Earth (m)

We can see that the apparent brightness depends on the distance between the Earth and the star, and it obeys the inverse-square law (Fig 3.2c), i.e. $b \propto \frac{1}{d^2}$. The value of b decreases to a quarter if the distance d is doubled, i.e. a star appears fainter if it is further away from us.

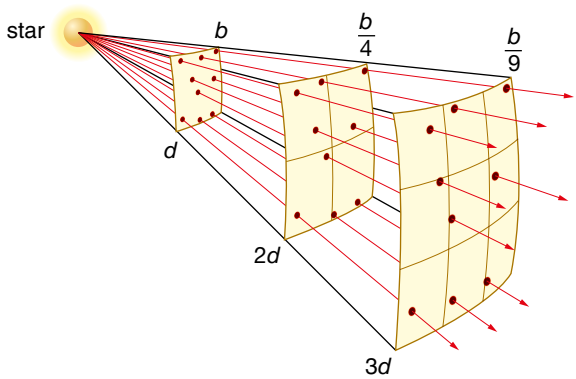


Fig 3.2c *The inverse-square law. Apparent brightness varies inversely as the square of distance to the star.*

→ This is a Video Quiz about an experiment verifying the inverse-square law of intensity.

apparent brightness 視亮度

Video

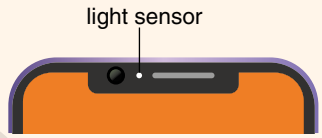
→ This video demonstrates the Phone experiment.

Phone experiment

→ Worksheet is available on OUP web.

Verifying the inverse-square law of apparent brightness

Smartphones are usually equipped with a light sensor which is used to adjust the brightness of the screen automatically according to the environment. The sensor is usually located at the top of the screen.



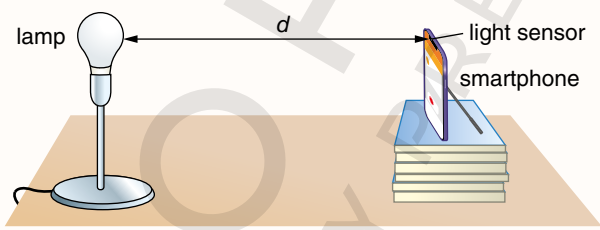
What to do

You may search the app on the Internet.

Light meter app

1 Open a light meter app that allows you to obtain the light sensor readings.

2 Place a lamp at a distance d away from the light sensor. Ensure that the sensor faces the lamp normally and is at the same height as the lamp.



Remove any lamp shade.

3 Darken the room and switch on the lamp. The light sensor will measure the brightness of the lamp. Record different values of d and the corresponding light meter reading L.

4 Calculate the values of $\frac{1}{d^2}$. Plot a graph of L against $\frac{1}{d^2}$.

What you find

1 How does the graph look like?

2 What is the relationship between L and $\frac{1}{d^2}$?

Example 2 Luminosity and apparent brightness of Vega

The distances of the Sun and Vega from the Earth and their luminosities are listed below. Estimate the apparent brightness of Vega as seen from the Earth. Given: $L_\odot = 3.84 \times 10^{26}$ W, 1 AU $= 1.50 \times 10^{11}$ m

Star	Distance from the Earth / AU	Luminosity
Sun	1	$L_\odot$
Vega	1.58×10^6	$40L_\odot$

Solution

$$b = \frac{L}{4\pi d^2} = \frac{40(3.84 \times 10^{26})}{4\pi[(1.58 \times 10^6)(1.50 \times 10^{11})]^2} = 2.18 \times 10^{-8} \text{ W m}^{-2}$$

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) If star A appears brighter than star B in the sky, A must be closer to us than B. (T/F)

(b) The luminosity of a star depends on how far away the observer is. (T/F)

(c) A star appears brighter in the sky must have a higher luminosity. (T/F)

2 Complete the table below.

	Star	L / W	b / W m^{-2}	d
(a)	P	5×10^{26}		2.5×10^{6} ly
(b)	Q	2×10^{28}	0.24	m
(c)	R		3×10^{-5}	12 pc

DSE goal

Realize magnitude as a measure of brightness of celestial bodies.

Distinguish between apparent magnitude and absolute magnitude.

DSE exam

15(2)Q1.3, 1.7, 16(2)Q1.7, 17(2)Q1.5, 18(2)Q1.6, 20(2)Q1.4, 21(2)Q1.4, Q1.5, 22(2)Q1.6, 23(2)Q1.7

2 Stellar magnitude

Astronomers measure and compare the brightness of celestial bodies in a *magnitude scale*.

a Apparent magnitude

In a magnitude scale, **apparent magnitude**, denoted by m, is used instead of apparent brightness.

The apparent magnitude m measures the brightness of celestial bodies as seen by an observer on Earth.

The **brighter** a celestial body appears, the **smaller** its apparent magnitude. The apparent magnitudes of some celestial bodies and the detection limits of some instruments are shown below (Fig 3.2d). Note that apparent magnitude can be negative.

Apparent magnitude and the absolute magnitude defined here are called the bolometric magnitudes (熱星等). They include contribution of energy from the whole EM spectrum. If one only considers the visible spectrum, the quantities are visual magnitudes (目視星等).

The more negative the magnitude, the brighter the object. ▶

Sirius (天狼星), in the constellation of Canis Major (大犬座), is actually a binary star system consisting of Sirius A and Sirius B with apparent magnitudes of –1.46 and 8.44 respectively. Sirius is a bluish white star that can be observed in Hong Kong in winter.

Polaris (北極星), in the constellation of Ursa Minor (小熊座), is actually a multiple star system. It is located within one degree of the north celestial pole today and is therefore the current north star. However, Polaris will not always be the north star. As the Earth slowly wobbles on its rotational axis in a 26 000-year cycle. Polaris can be seen in Hong Kong all year round.

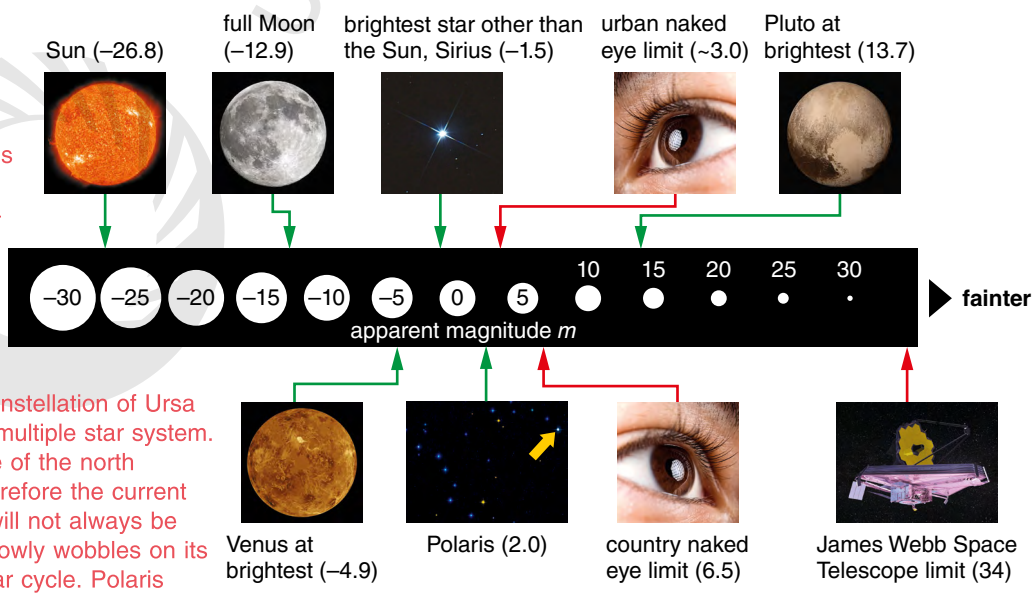


Fig 3.2d The apparent magnitude scale.

magnitude scale 星等標度 apparent magnitude 視星等

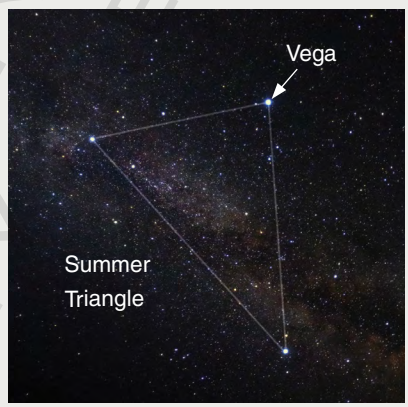
As mentioned above, the apparent brightness of a star depends on its luminosity and its distance from the Earth. Similarly,

the apparent magnitude *m* of a star depends on its luminosity and its distance from the Earth.

Know more

The magnitude system

In 1856, Oxford astronomer *Norman Pogson* suggested that a star's magnitude should be defined in terms of the star's apparent brightness, so that two astronomers could agree on exactly how bright a star was. He defined his scale such that an increase of 5 magnitude numbers meant a 100-fold decrease in brightness. Astronomers once chose Vega as the reference point of $m = 0$. The magnitude scale was extended to brighter bodies by giving them negative numbers.



✎ Human eye responds to light logarithmically. Therefore, the magnitude system is actually based on the response of the human eye.

b Absolute magnitude

It is difficult to tell which star emits more light by using apparent magnitude because the stars are at different distances from us. To make the comparison easier, astronomers compare how bright the stars would appear if they were 10 pc away. This is called the **absolute magnitude**, which is denoted by M.

The absolute magnitude *M* of a star is defined as the apparent magnitude of the star if it were located at a distance of 10 pc from the Earth.

Therefore,

the absolute magnitude *M* depends only on a star's luminosity and is independent of its distance from the Earth.

The **smaller** the absolute magnitude, the **more luminous** the star is.

Know more

More about *m* and *M*

It is defined that Δm of 5 corresponds to a difference of 100 times in apparent brightness. A star with m of 1 is 2.512 times brighter than a star with m of 2. Mathematically,

if $m_A - m_B = 5$,

$$\frac{b_A}{b_B} = 100$$

if $m_A - m_B = 1$,

$$\frac{b_A}{b_B} = \sqrt[5]{100} \approx 2.512$$

Similarly, ΔM of 5 corresponds to a difference of 100 times in luminosity.

Norman Pogson 普森　　absolute magnitude 絕對星等

Altair（牛郎星/河鼓二）is a white star in the constellation of Aquila（天鷹座）and is the twelfth brightest star in the night sky. Altair and two other bright stars, Vega（織女星）in Lyra（天琴座）and Deneb（天津四）in Cygnus（天鵝座）, form the Summer Triangle (see the photo in Know more on p.108), a prominent sign in the summer night sky of the northern hemisphere.

Example 3 Relating magnitudes with luminosities and apparent brightness

The following table lists the luminosities and apparent brightness of the Sun and the star Altair.

Star	Luminosity / W	Apparent brightness / W m^{-2}
Sun	3.84×10^{26}	1358
Altair	4.07×10^{27}	1.29×10^{-8}

(a) (i) Which of them gives out more energy into space per unit time? Explain.

(ii) Which of them looks brighter as observed from the Earth? Explain.

(b) Hence, compare their (i) absolute magnitudes and (ii) apparent magnitudes.

Solution

(a) (i) Since Altair has a greater luminosity, it gives out more energy into space per unit time.

(ii) Since the Sun has a higher apparent brightness, it looks brighter.

(b) (i) The absolute magnitude of Altair is smaller than that of the Sun.

(ii) The apparent magnitude of the Sun is smaller than that of Altair.

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) A star with an apparent magnitude of 0 appears dimmer than a star with an apparent magnitude of 1. (T/F)

(b) If two stars have the same apparent magnitude, the energy received per unit area per unit time from these two stars must be equal on Earth. (T/F)

(c) The absolute magnitude of the Sun is larger than its apparent magnitude. (T/F)

(d) The larger the absolute magnitude of a star is, the lower its luminosity is. (T/F)

2 The apparent magnitude m and the absolute magnitude M of some stars are listed below.

Star	W	X	Y	Z
m	0	7	2	−3
M	9	−2	3	−1

(a) Which star looks the brightest as observed from the Earth?

(b) Which star has the highest luminosity?

(c) Which star allows a telescope on Earth to collect the largest amount of energy per unit time from it?

(d) The limit of the naked eye is about magnitude 6.5. Which star(s) cannot be observed with the naked eye?

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C Relationship between apparent and absolute magnitudes

Two stars of the same apparent magnitude (m) can differ a lot in their absolute magnitudes (M). In Figure 3.2e, star A has a greater luminosity than star B but is further away from the Earth. However, both stars A and B seem to have the same brightness in the night sky.

Therefore, without knowing the distances of the two stars to the Earth, we cannot compare their luminosities by their apparent brightness.

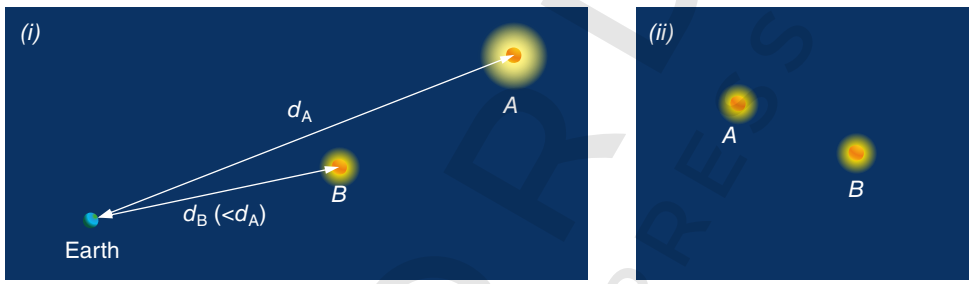


Fig 3.2e *(i) Star A has a smaller absolute magnitude and is further away from the Earth. (ii) Stars A and B have the same apparent magnitude in the sky.*

Table 3.2a shows the apparent and absolute magnitudes of some stars.

Star	Apparent magnitude m	Absolute magnitude M	Distance from the Earth
Sun	−26.8	4.83	4.85×10^{-6} pc (1 AU)
Sirius	−1.47	1.42	2.64 pc
Canopus	−0.72	−5.61	95 pc

Sirius is the brightest star in the night sky while Canopus is the second brightest.

Table 3.2a *Apparent and absolute magnitudes of some stars.*

Canopus（南極老人星/老人星/南極星）is a yellowish-white star in the southern constellation of Carina（船底座）. The star appears in the southern sky of Hong Kong in winter.

We could estimate the distance of a star by comparing its M and m (Fig 3.2f).

1. If $m < M$ (e.g. Sirius), the star looks brighter than when it were at 10 pc, so it is less than 10 pc away from us.
2. If $m = M$, the star is 10 pc away from us.
3. If $m > M$ (e.g. Canopus), the star looks fainter than when it were at 10 pc, so the star is more than 10 pc away from us.

The larger the difference between M and m, the larger the difference between 10 pc and the actual distance to the star. In fact, m and M are related by the equation

$$m - M = 5 \log\left(\frac{d}{10}\right)$$

where d is the distance to the star in parsec.

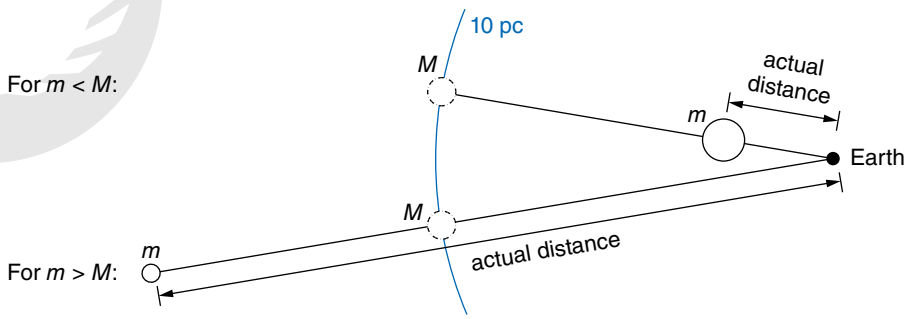


Fig 3.2f *The relationship between m, M and the distance of a star.*

Note that luminosity and absolute magnitude describe a property of the star and are independent of the distance of the star from the Earth. On the other hand, apparent brightness and apparent magnitude depend on how far the star is from the Earth. Their relationships are summarized in Figure 3.2g.

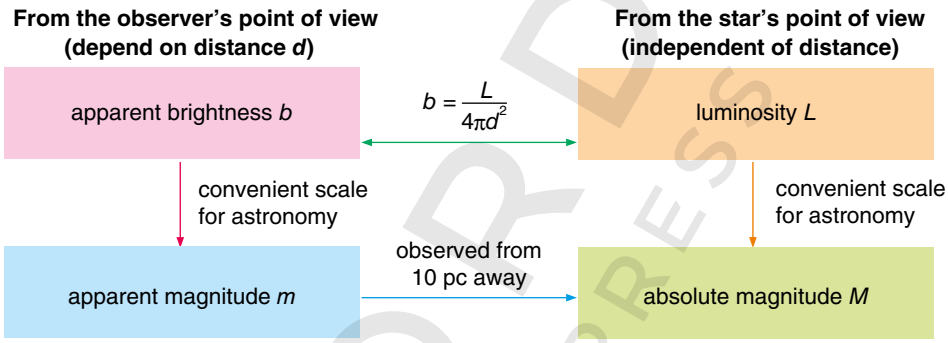


Fig 3.2g *The relationships between apparent brightness, luminosity, apparent and absolute magnitudes.*

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Two stars having the same absolute magnitude are at the same distance from the Earth. (T/F)

2 *True or false:* Fomalhaut is a bright star with an apparent magnitude of 1.16. It would appear brighter if it is at 10 pc away from the Earth.

(a) The absolute magnitude of Fomalhaut is larger than 1.16. (T/F)

(b) The distance between Fomalhaut and the Earth is larger than 10 pc. (T/F)

Learn by practice 2

1 The apparent magnitudes m and the absolute magnitudes M of some stars are listed in Table a.

Star	m	M
X	1	−1
Y	−0.5	−1
Z	−0.5	4

Table a

(a) Are Y and Z the same distance away from the Earth?

(b) Which star(s) is/are further than 10 pc away from the Earth?

2 The apparent magnitudes m and the absolute magnitudes M of two stars X and Y are listed in Table b.

Star	m	M
X	2.8	4.7
Y	5.4	3.2

Table b

(a) Which star is within 10 pc away from the Earth? Explain.

(b) Which star has a larger luminosity?

(c) If X is located at the position of Y instead, how will its brightness change as seen from the Earth? Explain.

☆ HKDSE 2016 Paper 2 Q1.7

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Apparent magnitude and luminosity

Stars P and Q have the same apparent magnitude. The parallax of P is only half of that of Q.

Find $\frac{\text{luminosity of } P}{\text{luminosity of } Q}$.

A $\frac{1}{4}$ B $\frac{1}{2}$

C 2 D 4

Solution

As $d = \frac{1}{p}$, when the parallax of P is only half of that of Q, the distance d to P is twice that of Q.

Common mistake ⚠
Students may fail to relate the parallax with the distance to the star.

P and Q have the same apparent magnitude, this means they have the same apparent brightness b. From $b = \frac{L}{4\pi d^2}$,

Common mistake ⚠
Students may fail to relate the apparent magnitude with the apparent brightness and hence the luminosity.

$$\frac{L_P}{4\pi {d_P}^2} = \frac{L_Q}{4\pi {d_Q}^2}$$

$$\frac{L_P}{L_Q} = \frac{{d_P}^2}{{d_Q}^2} = \frac{(2d)^2}{d^2} = 4$$

∴ The answer is D.

Revision 3 Q16 (p.148)

Exercise 3.2

(For Q1–3.) The apparent magnitudes m and absolute magnitudes M of four stars are shown in Table a.

Star	P	Q	R	S
m	0	−0.5	2	−2
M	−0.5	2	−1.5	−2.5

Table a

1 Which star appears the brightest as seen from the Earth?

A P B Q

C R D S

2 If all four stars are placed at the same distance from the Earth, which star appears the brightest as seen from the Earth?

A P

B Q

C R

D S

★**3** Which star is within 10 pc away from the Earth?

A P

B Q

C R

D S

★**4** Stars P and Q have the same luminosity. Which of the following statements must be correct?

(1) They have the same apparent magnitude.

(2) They have the same absolute magnitude.

(3) They give out an equal amount of electromagnetic energy into space every second.

A (2) only

B (3) only

C (1) and (3) only

D (2) and (3) only

★**5** Vega has apparent and absolute magnitudes 0.03 and 0.58 respectively. Which of the following statements about Vega is/are correct?

(1) It is not visible to the naked eye.

(2) Its luminosity is very close to zero.

(3) Its distance from the Earth is smaller than 10 pc.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

6 John looks at star A for one second and receives 3×10^{-10} J of energy from it. His pupil is 2.9 mm in radius (Fig a).

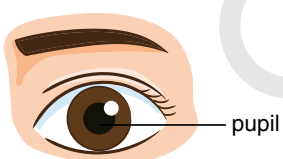


Fig a

(a) Suppose the apparent brightness of star B is 2×10^{-8} W m^{-2}, which star appears brighter?

(b) If star B is 130 pc away from us, what is the power given out by star B?

7 The apparent magnitudes of stars P, Q and R are 0.5, 2 and 8 respectively.

(a) Which star(s) is/are visible to the naked eye?

(b) Which star appears the brightest in the sky?

(c) Explain qualitatively the relationship between apparent magnitude and luminosity of a star.

★**8** Polaris (Fig b) has an apparent magnitude of 1.97 and an absolute magnitude of –3.6.



Fig b

(a) Write down the definition of absolute magnitude.

(b) What conclusion can you draw about the distance of Polaris from the Earth? Explain your answer.

(c) Suppose another star has a luminosity greater than that of Polaris. Is its absolute magnitude larger or smaller than Polaris?

★**9** Consider the stars in Table b.

Star	X	Y	Z
Apparent magnitude	11	–0.5	–0.5
Absolute magnitude	6	6	–4.5

Table b

(a) Which stars have the same brightness as observed from the Earth?

(b) Star X and Y have the same absolute magnitude. What does this mean? Why do they have different apparent magnitude?

(c) Which of them is the closest to the Earth? Explain briefly.

★**10** Given the apparent brightness of the Sun observed from the Earth is 1358 W m^{-2}.

(a) Find the apparent brightness of the Sun observed from Neptune, which is 30.1 AU away from the Sun.

(b) How many times dimmer would the Sun be if observed from 10 pc away compared to that from the Earth?

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3.3 Stellar Spectrum and H-R diagram

When metals are heated to high temperatures, they glow yellow and even white (Fig 3.3a). Stars also shine in different colours (Fig 3.3b). Are the colours of stars related to temperature just like the heated metals?



Fig 3.3a *Heated metal with different colours.*



Fig 3.3b *Stars with different colours.*

Let us study the light (electromagnetic radiation) emitted by stars.

1 Stellar spectrum

a Blackbody radiation

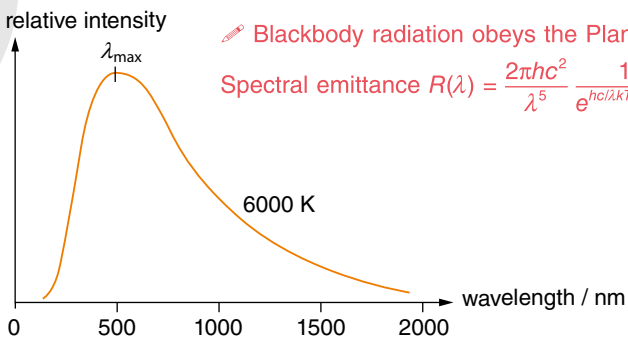
Some Ss may wrongly conclude that a blackbody does not emit any radiation simply because it absorbs all radiation and reflects none.

A blackbody absorbs all EM radiation, including all colours of light, falling on it. Therefore, it appears black and is named 'blackbody'.

▶ A star behaves like a **blackbody**, which is a perfect absorber and emitter of electromagnetic radiation. The radiation emitted by a blackbody is called **blackbody radiation**.

Both intensity *I* and apparent brightness *b* are power per unit area. In this book, *I* refers to the power **emitted** per unit area by a body, while *b* refers to the power **received** per unit area on Earth.

▶ We use **intensity** to describe how strong the electromagnetic radiation emitted by a body is. It is the power emitted per unit area. Figure 3.3c shows the relative intensity of different wavelengths of radiation emitted by a blackbody. This is called a blackbody radiation curve.



Blackbody radiation obeys the Planck's law.

Spectral emittance $R(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

Fig 3.3c *A blackbody radiation curve at a surface temperature of 6000 K.*

Simulation

→ This simulation shows how the radiation curve changes with the surface temperature of a blackbody. It is available on OUP web in PC version (.exe) only.

blackbody 黑體　　blackbody radiation 黑體輻射

Note that:

1 The curve is continuous. This shows that a blackbody emits a continuous spectrum of radiation.

2 The curve has a peak at a specific wavelength λ_{max}. Radiation of this wavelength has the highest intensity.

DSE goal

Describe the effect of surface temperature on the colour and luminosity of a star using blackbody radiation curves.

DSE exam

17(2)Q1.6, 19(2)Q1.7, 20(2)Q1.8

b Surface temperature and blackbody radiation

The intensity of radiation emitted by a blackbody depends only on its **surface temperature**. Figure 3.3d shows blackbody radiation curves at different surface temperatures.

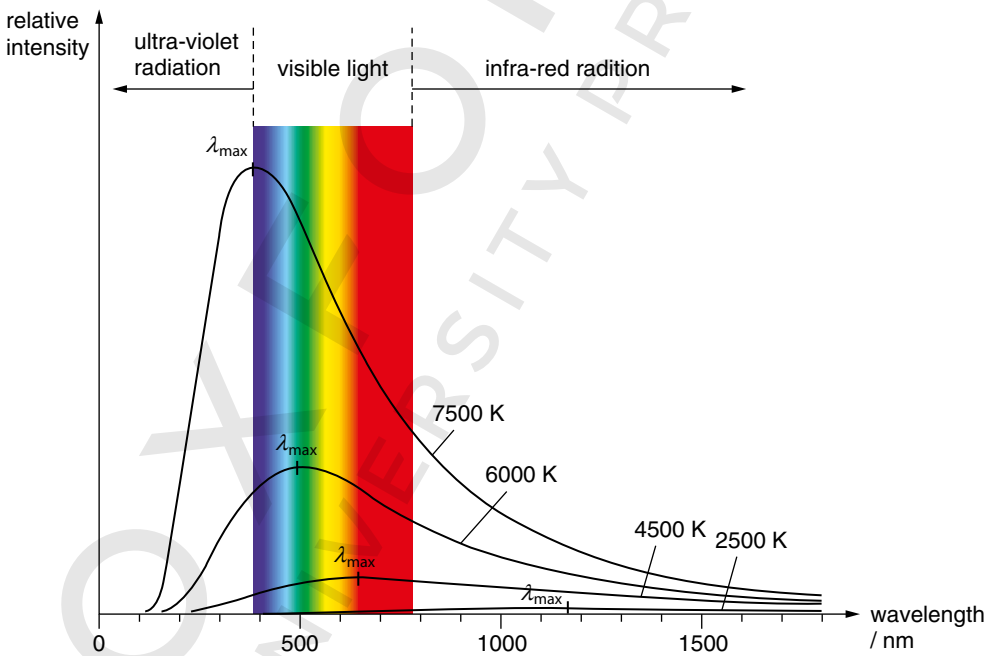


Fig 3.3d Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.

Let us see the features of the blackbody radiation curves in details.

i Peak wavelength

Curves at different surface temperatures T have different peak wavelengths λ_{max}. The peak shifts towards shorter wavelengths as T increases, i.e.,

$$T\uparrow \Rightarrow \lambda_{max}\downarrow$$

In fact, λ_{max} is inversely proportional to the temperature of the radiation curve. ▶

A star emits electromagnetic radiation like a blackbody. The relative intensity of visible light determines the colour of the star. The radiation emitted by a star having a higher surface temperature peaks at a shorter wavelength and the star looks bluer. Stars have different surface temperatures, so they have different colours.

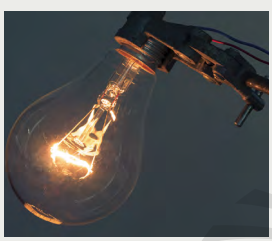
The Sun has surface temperature at around 6000 K. Therefore, the radiation curve of the Sun peaks at the blue-green portion of the visible spectrum as shown in Figure 3.3d. ▶

Note that although the 6000-K curve peaks at green, the Sun does not look green. The 6000-K curve shows roughly equal intensities of different colours of visible light. Therefore, the Sun looks white.

Know more

Incandescent lamps

The filament in an incandescent lamp behaves like a blackbody. A 2500-K filament emits blackbody radiation that peaks at around 1160 nm, which is infra-red radiation. Since most radiation emitted is not visible light, the energy efficiency of the lamp is very low.



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Figure 3.3e shows the colours of stars perceived. From cold to hot, the possible colours of stars are red, orange, yellow, white, bluish-white and blue.

Know more

No green or violet stars observed?

We seldom see stars in green or violet in the sky. Why?

▸ p.156

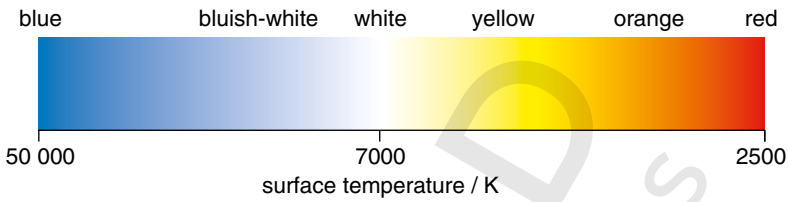


Fig 3.3e *The colours of stars with different surface temperatures.*

A cooler star appears redder and a hotter star appears bluer.

Therefore, we can estimate the surface temperature of a star from its colour observed.

ii Total power radiated

The blackbody radiation curves at different temperatures do not intersect with each other. For two blackbodies of the same size, the hotter one emits more radiation at **all** wavelengths as shown in Figure 3.3f.

The area under each curve represents the total power radiated per unit surface area of the blackbody. Figure 3.3f clearly shows that the total power per unit surface area radiated by a 6000-K star, i.e. the blue and red area, is more than that by a 4500-K star, i.e. the red area.

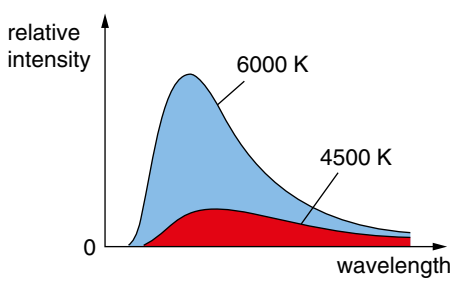


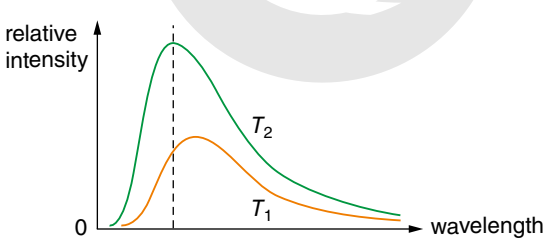
Fig 3.3f *Blackbody radiation curves at 6000 K and 4500 K.*

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Refer to the spectra of radiation from a blackbody at two different temperatures, T_1 and T_2, as shown below. Determine whether the following statements are true.



(a) T_2 is larger than T_1. (T/F)

(b) The body appears redder at T_2. (T/F)

(c) The body gives out more energy per unit time at T_2. (T/F)

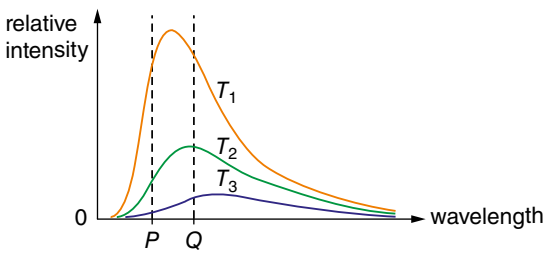
2 *True or false:*

(a) Stars emit light within the visible spectrum only. (T/F)

(b) The peak of the blackbody radiation curve of a star is always located within the visible spectrum. (T/F)

Learn by practice 3

1 The figure below shows the radiation curves from different stars. P and Q denote the lower and upper wavelength limits of the visible spectrum respectively. T_1, T_2 and T_3 are temperatures of the respective radiation curves with one of them belonging to the Sun.



(a) Which curve belongs to the Sun?

(b) Which wavelength limit, P or Q, denotes the wavelength for red?

(c) Can we see the star with T_3? Explain why.

2 The radiation curve of blackbody A peaks at 400 nm. The radiation curve of another blackbody B, of the same size, peaks at 800 nm.

(a) Which blackbody gives out more radiation at 800 nm?

(b) Which blackbody has a higher surface temperature?

(c) Which blackbody appears redder?

(d) Which blackbody gives out more radiation summed over the whole spectrum?

DSE goal

State Stefan's law and apply it to derive the luminosity $L = 4\pi R^2\sigma T^4$ for a spherical blackbody.

DSE exam

12(2)Q.1, 14(2)Q.1(a)–(b)(ii), 22(2)Q1.5, 23(2)Q.1(a)–(b)

C Stefan's law

Stefan's law describes the relationship between the total power radiated per unit surface area J of a blackbody and its surface temperature T. It states that the total energy radiated per unit surface area of a blackbody in unit time is proportional to the fourth power of its temperature. Mathematically, it can be expressed as

total power radiated per unit surface area by blackbody (W m^{-2}) — Stefan constant = 5.67×10^{-8} W m^{-2} K^{-4}

$$J = \sigma T^4$$

surface temperature of blackbody (K)

Stefan's law is also known as the Stefan–Boltzmann law. It only account for blackbodies, while luminosity and intensity give the total power radiated by any object.

where $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} is called the Stefan constant (or Stefan-Boltzmann constant), and T is measured in K. The unit of J is W m^{-2}.

For a star of luminosity L and surface area $4\pi R^2$ (where R is the radius of the star), Stefan's law can be rewritten as:

$$J = \frac{L}{\text{area}} = \frac{L}{4\pi R^2} = \sigma T^4$$

When applying Stefan's law to a star, J can be regarded as the intensity of the star.

Therefore, the luminosity of the star is:

luminosity of star (W or J s^{-1}) — Stefan constant = 5.67×10^{-8} W m^{-2} K^{-4}

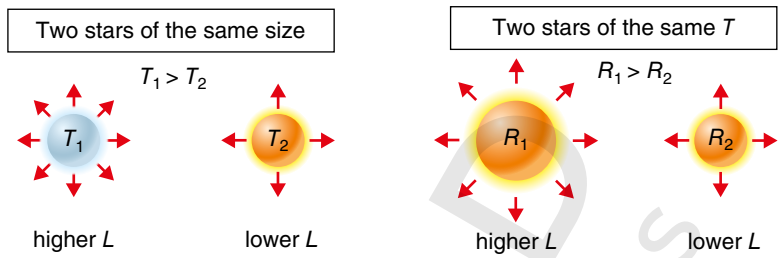
$$L = 4\pi R^2\sigma T^4$$

radius of star (m) — **surface temperature** of star (K)

Stefan's law is sometimes written as $P = \sigma AT^4$ for blackbodies, where P is the power emitted by the star (i.e. luminosity) and A is the surface area of the star.

Stefan's law 斯特藩定律

From the equation, we can see that the luminosity of a star depends on both its surface temperature and its radius (Fig 3.3g).



(i) A hotter star has a higher L if R of two stars are the same.

(ii) A larger star has a higher L if T of two stars are the same.

Fig 3.3g *The relationships between L, T and R of a star.*

The Stefan's law is useful for finding the radius of a star. See Example 4.

Arcturus (大角星) is an orange star in the constellation of Boötes (牧夫座). It is the fourth brightest star in the night sky.

Example 4 Arcturus

The table lists some observation data of star Arcturus.

Parallax *p*	88.78×10^{-3}″
Apparent brightness *b*	4.54×10^{-8} W m^{-2}
Surface temperature *T*	4300 K

The *p* and *b* of a star can be measured directly while *T* is obtained by studying the black-body radiation curve of the star.

(a) Estimate the radius of Arcturus. Take $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

(b) Arcturus is 180 times more luminous than the Sun. Find the ratio of the apparent brightness of Arcturus to that of the Sun.

Solution

1 pc = 3.09×10^{16} m

(a) By $d = \dfrac{1}{p}$, $d = \dfrac{1}{88.78 \times 10^{-3}} = 11.26$ pc $= 11.26 \times 3.09 \times 10^{16}$

$= 3.48 \times 10^{17}$ m

By $b = \dfrac{L}{4\pi d^2}$, $L = 4\pi d^2 b = 4\pi(3.48 \times 10^{17})^2(4.54 \times 10^{-8}) = 6.91 \times 10^{28}$ W

By $L = 4\pi R^2 \sigma T^4$,

radius of Arcturus $= \sqrt{\dfrac{L}{4\pi\sigma T^4}} = \sqrt{\dfrac{6.91 \times 10^{28}}{4\pi \times (5.67 \times 10^{-8}) \times 4300^4}}$

$= 1.68 \times 10^{10}$ m

(b) By $b = \dfrac{L}{4\pi d^2}$,

$\Rightarrow b \propto \dfrac{L}{d^2}$

$\Rightarrow \dfrac{b_A}{b_\odot} = \left(\dfrac{L_A}{L_\odot}\right)\left(\dfrac{d_\odot}{d_A}\right)^2 = (180)\left(\dfrac{1.50 \times 10^{11}}{3.48 \times 10^{17}}\right)^2 = 3.34 \times 10^{-11}$

Tip

The Earth-Sun distance $d_\odot$ = 1 AU = 1.50×10^{11} m

Quick check 7

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) If star A has a surface temperature higher than star B, star A must have a higher luminosity than star B. (T/F)

(b) The power radiated per unit surface area by a blackbody at 2000 °C is 16 times that of a blackbody at 1000 °C. (T/F)

2 The peaks of the spectra of radiation from two stars of different sizes X and Y lie at the same wavelength. X looks brighter than Y. Determine whether the following statements are true.

(a) The apparent brightness of X and Y are the same. (T/F)

(b) The surface temperature of X and Y are the same. (T/F)

(c) The luminosity of X and Y may be the same. (T/F)

3 The luminosities L, the radii R and the surface temperatures T of some stars are shown below. Complete the table.

Take $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

	Star	L / W	R / m	T / K
(a)	P		6.96×10^{8}	5780
(b)	Q	4.50×10^{31}	6.20×10^{11}	
(c)	R	7.50×10^{31}		8500
(d)	S		1.41×10^{11}	8525

Learn by practice 4

1 The Earth receives solar radiation of power P_0 per unit area. Estimate the power of solar radiation received per unit area on Pluto which is 40 AU from the Sun. Express your answer in terms of P_0.

2 Vega is a star of surface temperature 9600 K. It is 25 ly away from the Earth. It has luminosity of 1.54×10^{28} W, which is 40 times more luminous than the Sun.

(a) Estimate the radius of Vega. Take $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

(b) Find the ratio of the apparent brightness of Vega to that of the Sun. Take 1 ly = 63 100 AU.

DSE goal

Realize the existence of spectral lines in the spectra of stars.

DSE exam

12(2)Q1.8, 18(2)Q1.4

d Spectral lines

We have assumed that a star behaves like a blackbody by far. In fact, stars deviate slightly from a blackbody when we examine their spectra carefully.

When the radiation from a blackbody passes through a prism or a plane transmission grating, a continuous spectrum is formed (Fig 3.3h).



Fig 3.3h *The blackbody radiation spectrum is continuous.*

Cross-link

You may refer to Book E2 Ch 2 for more details about the formation of different spectra.

Cross-link

You may refer to Book 3B Ch 6 for more details about the formation of spectrum through plane transmission grating.

The same occurs in the radiation from stars except that there are dark lines in their spectra (Fig 3.3i). This kind of spectrum is called an **absorption spectrum** and the dark lines are called **absorption lines**. They are one kind of **spectral lines**.

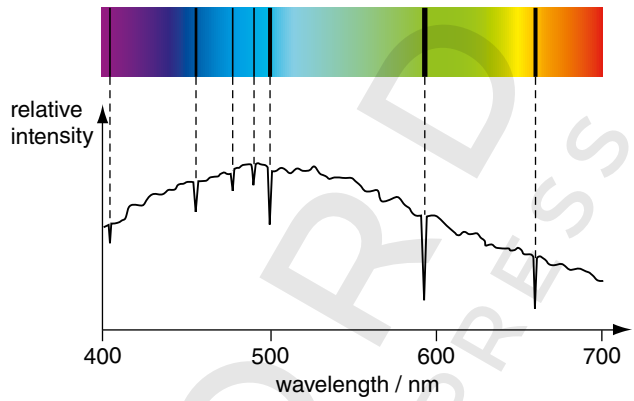


Fig 3.3i *Dark lines in the spectrum of sunlight. The corresponding radiation curve shows the reduced intensities at wavelengths of the dark lines.*

Experiment 4a

Observing spectra (p.163)

Video

→ This video demonstrates Expt 4a.

Absorption lines are formed when radiation of certain wavelengths is absorbed by the gas at the star's outer surface. The gas then re-radiates the radiation in random directions. As a result, in the observed starlight, the intensity of those wavelengths is reduced and dark lines are formed on the spectrum (Fig 3.3j).

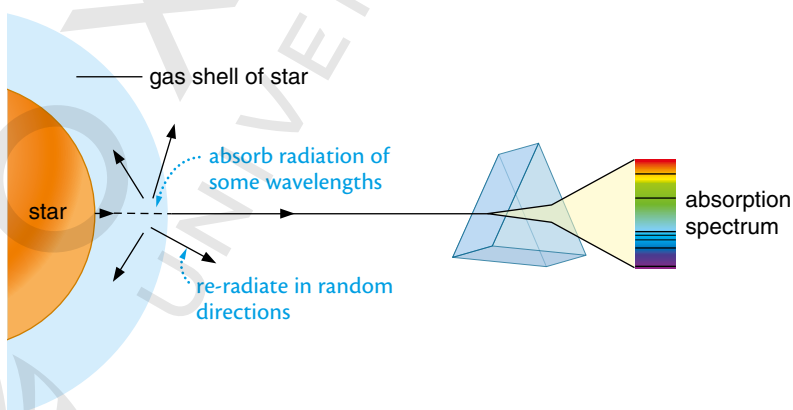


Fig 3.3j *Formation of absorption spectrum.*

The absorption spectra give us a lot of information about the stars.

1 Each element has its own characteristic pattern of spectral lines. Therefore, by examining the spectrum, we can find what chemical elements exist in the gas at the star's outer surface.

✎ Each element has a unique set of energy levels, which corresponds to a unique set of absorption lines.

2 The absorption spectrum looks different at different temperatures. Therefore, the surface temperature of a star can be deduced.

Stars are classified according to their spectral characteristics and hence their surface temperatures. Details will be given in the next section on p.121. ▶

3 Spectral lines can be used to determine how fast the star is moving towards or away from us.

This will be further discussed in Ch 3.4. ▶

absorption spectrum (pl. spectra) 吸收光譜　absorption line 吸收譜線　spectral line 光譜線

Google Forms Microsoft Forms

Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 The atmosphere of a blackbody contains hydrogen and helium gases. Consider the spectrum of the blackbody. Determine whether the following statements are true.

(a) The absorption lines of hydrogen and helium appear at different wavelengths in the spectrum. (T/F)

(b) The absorption spectrum contains one dark line for each element. (T/F)

(c) The absorption lines show that the intensity of the radiation at the corresponding wavelengths is zero. (T/F)

2 *True or false:* The chemical composition of a star's outer surface can be deduced by analysing its absorption spectrum. (T/F)

3 The absorption spectrum of star *X* contains hydrogen absorption lines. Determine whether the following statements are true.

(a) Star *X* is mainly composed of hydrogen. (T/F)

(b) There is hydrogen gas in the outer atmosphere of star *X*. (T/F)

DSE goal

State major spectral classes and relate them to the surface temperature of stars.

DSE exam

14(2)Q.1(b)(iii), 19(2)Q1.6

2 Spectral classes of stars

Astronomers classify stars into seven major **spectral classes** according to their surface temperature, with O being the hottest and M being the coolest (Table 3.3a).

You may find it easier to memorize the order of the class symbols this way: **O**h, **B**e **A** **F**ine **G**irl/**G**uy, **K**iss **M**e.

Class	Surface temperature	Colour	Example
O	25 000–50 000 K	blue	Meissa
B	10 000–25 000 K	bluish-white	Spica
A	7400–10 000 K	white	Vega
F	6000–7400 K	yellowish-white	Canopus
G	5000–6000 K	yellow	Sun
K	3500–5000 K	orange	Arcturus
M	< 3500 K	red	Betelgeuse

Table 3.3a *Surface temperatures, colours and examples of the major spectral classes.*

Our Sun is a G star with a surface temperature of 5800 K. Alpha Centauri A is also a G star. Its surface temperature and colour should be similar to that of the Sun.

Know more

The Harvard spectral classification

This stellar classification system is also called the Harvard spectral classification. Do you know how it was developed?

▸ p.155

spectral class 光譜型

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Arrange the following classes of star in ascending order of their surface temperature.

A O M B K

2 In what spectral class does the Sun belong?

3 What is the surface temperature range of a star in the same class as the Sun?

A 25 000–50 000 K
B 6000–7400 K
C 5000–6000 K
D 3500–5000 K

DSE goal

Represent information of classification for stars on the H-R diagram according to their luminosities and surface temperatures.

DSE exam

12(2)Q1.7, 17(2)Q1.7, 18(2)Q1.7, 20(2)Q.1(a)–(c), 22(2)Q1.8

3 Hertzsprung-Russell diagram

If we plot a graph of stars' luminosities against their surface temperatures, we obtain the **Hertzsprung-Russell diagram**, or **H-R diagram** for short (Fig 3.3k).

The H-R diagram is named after its originators, *Ejnar Hertzsprung* and *Henry Norris Russell*. They created the diagram in about 1910.

A simplified H-R diagram is usually drawn in dots and without indicating stars' colours and sizes.

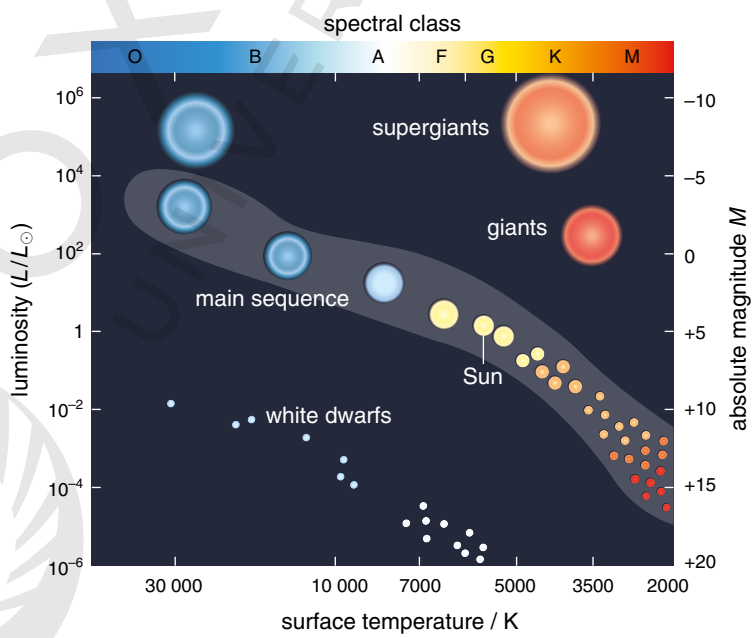


Fig 3.3k *The H-R diagram. (The sizes of stars are not drawn to scale.)*

In the H-R diagram, the vertical axis can be luminosity L, absolute magnitude M or luminosity relative to that of the Sun $\left(\frac{L}{L_\odot}\right)$. The horizontal axis can be surface temperature T or spectral class. Note that the surface temperature is **decreasing from the left to the right**.

Hertzsprung-Russell diagram (H-R diagram) 赫羅圖 Ejnar Hertzsprung 赫茲普龍
Henry Norris Russell 羅素

a Groups of stars

In the H-R diagram, the stars are not evenly distributed but seem to form groups. There are three main groups (Table 3.3b).

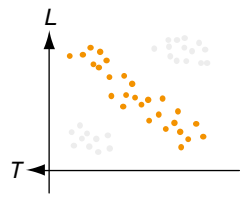
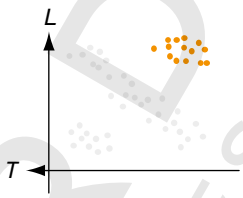
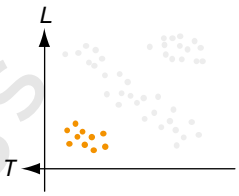
Group	Main sequence	Giant / red giant	White dwarf
Position in H-R diagram	 Diagonally from upper left to lower right	 Upper right, above the main sequence	 Lower left, below the main sequence
Property	• In general, stars with higher surface temperatures are more luminous • Various luminosities, surface temperatures and radii	• high luminosity • low surface temperature • large radius	• low luminosity • high surface temperature • small radius ✎ A typical white dwarf has half solar mass but only about 0.01 solar radius.
Colour	Varies	Usually red	Usually bluish-white
Remark	About 90% of stars belong to this group. They are called **main sequence stars**.	Larger stars are called **supergiants**.	Usually about the size of the Earth.
Example	Sun	Betelgeuse	Sirius B

Table 3.3b *Different groups of stars in the H-R diagram.*

Usually, the greater the mass of a main sequence star, the larger its radius and the higher its surface temperature, hence, the higher its luminosity. ▶

✎ Main sequence stars fuse hydrogen into helium within their cores. They have mass ranging from 0.1 to 200 solar mass. The Sun is a medium-sized and relatively normal main sequence star.

✎ A red giant is a dying star in the final stages of the star life cycle. Typical red giants have radii ranging from about 10 to 100 solar radii.

DSE goal

Use H-R diagram and Stefan's law to estimate the relative sizes of stars.

DSE exam

13(2)Q1.7, 16(2)Q.1, 17(2)Q1.8, 19(2)Q1.4

b Relative sizes of stars

From the H-R diagram and the equation $L = 4\pi R^2\sigma T^4$, we can compare the size of stars. Consider the luminosities of a star and the Sun.

$L = 4\pi R^2\sigma T^4$ (for a star)

and

$L_\odot = 4\pi R_\odot{}^2\sigma T_\odot{}^4$ (for the Sun)

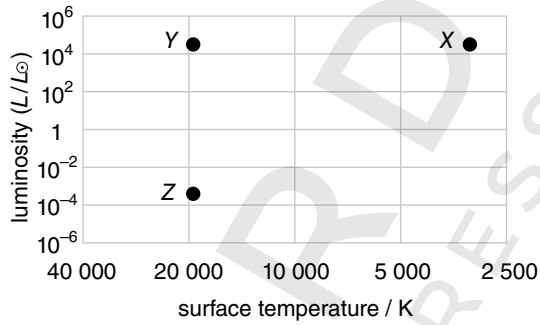
Therefore,

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2\left(\frac{T}{T_\odot}\right)^4$$

main sequence 主序　main sequence star 主序星　giant 巨星　red giant 紅巨星
supergiant 超巨星　white dwarf 白矮星

Example 5 Sizes of stars

Arrange stars X, Y and Z in the figure below in ascending order of their sizes. Explain briefly.



Solution

From $L \propto R^2T^4$, Z has the same surface temperature but a lower luminosity than Y, so Z has a smaller radius than Y.

While X has the same luminosity but a lower surface temperature than Y, so X has a larger radius than Y.

$\therefore Z < Y < X$

Example 6 Comparing Altair and the Sun

Star Altair has a surface temperature of 7350 K and a luminosity of $10.6L_\odot$. The Sun is a class G star with a surface temperature of 5780 K.

(a) Deduce whether the spectral class of Altair is class A or M.

(b) Indicate the position of Altair relative to the Sun in an H-R diagram.

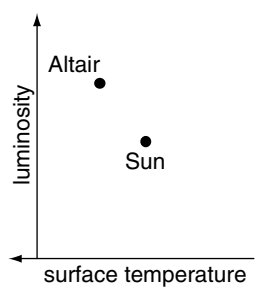
(c) Estimate the radius of Altair in terms of solar radius $R_\odot$.

(d) Does Altair or the Sun give out more red light per unit surface area? Explain.

Solution

(a) Class A. As Altair has a higher surface temperature than the Sun.

(b) See the H-R diagram on the right. Altair is hotter than the Sun. It should be to the left of the Sun in the diagram. Also, Altair is more luminous than the Sun. Therefore, it should be at the upper left above the Sun.



(c) By $\frac{L_A}{L_\odot} = \left(\frac{R_A}{R_\odot}\right)^2 \left(\frac{T_A}{T_\odot}\right)^4$,

$$R_A = \sqrt{\frac{L_A}{L_\odot}} \times \left(\frac{T_\odot}{T_A}\right)^2 R_\odot = \sqrt{10.6} \times \left(\frac{5780}{7350}\right)^2 R_\odot = 2.01 R_\odot$$

Star behaves like a blackbody. ▶

(d) Altair gives out more red light per unit surface area than the Sun. Altair has a higher surface temperature, so it emits more radiation at all wavelengths including red light than the Sun.

Quick check 10

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false*: The spectral class of a star can be deduced from the position of the star in the H-R diagram. (T/F)

2 Which of the following is a feature of red giants compared to main sequence stars?

A Higher surface temperature

B Further away from the Earth

C Larger radius

D Lower luminosity

3 Label the different groups of stars in the H-R diagram below (Fig a).

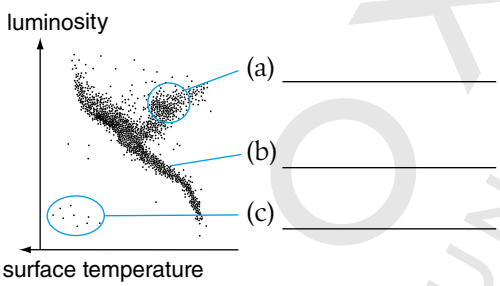


Fig a

4 The positions of four stars *W*, *X*, *Y* and *Z* are shown in the H-R diagram (Fig b). Determine whether the following statements are true.

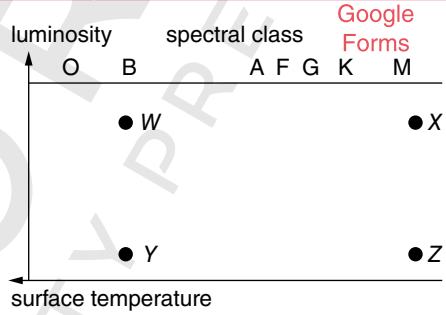


Fig b

(a) Red light has higher intensity than other colours of light in the spectrum of *X*. (T/F)

(b) Blue light has higher intensity than other colours of light in the spectrum of *W*. (T/F)

(c) Intensity ratio of red light to other colours of light is larger in the spectrum of *Z* than that of *Y*. (T/F)

(d) Luminosity of *Y* > luminosity of *X* (T/F)

(e) *X* has the largest radius among the four stars. (T/F)

(f) Radius of *W* > radius of *Y* (T/F)

(g) Radius of *Y* > radius of *Z* (T/F)

Learn by practice 5

1 Star *X* has the same surface temperature as the Sun but its luminosity is just half of that of the Sun.

(a) Indicate the position of *X* relative to the Sun in the H-R diagram.

(b) Estimate the radius of star *X* in terms of solar radius $R_\odot$.

(c) Compare the absolute magnitude of *X* with the Sun.

2 Star *Y* has a radius of $3R_\odot$ and a luminosity of $12L_\odot$.

(a) Estimate the surface temperature of *Y*, in terms of solar surface temperature $T_\odot$.

(b) Estimate the luminosity of *Y* in watts.

Given: $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}, $R_\odot = 6.96 \times 10^8$ m, $T_\odot = 5780$ K

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Exercise 3.3

(For Q1–3.) The radiation curves of blackbodies X, Y and Z peak at 450 nm, 720 nm, and 650 nm respectively.

1 Which blackbody has the highest surface temperature?

A X B Y
C Z D Cannot be determined

★**2** Which blackbody emits the most infra-red radiation per unit area?

A X B Y
C Z D Cannot be determined

★**3** Which blackbody has the highest luminosity?

A X B Y
C Z D Cannot be determined

★**4** The relative luminosities and radii of stars X, Y and Z are shown in Table a.

Star	X	Y	Z
Luminosity	$8L_\odot$	$4L_\odot$	$2L_\odot$
Radius	$2R_\odot$	$R_\odot$	$0.5R_\odot$

Table a

Which star has the highest surface temperature?

A X
B Y
C Z
D They have the same temperature.

5 The radiation curves of blackbodies A and B peak at 200 nm (ultra-violet radiation) and 650 nm (red light) respectively.

(a) Which kind of EM wave does each of the objects mainly emit?

(b) Which blackbody is hotter?

(c) Figure a shows the radiation curve of blackbody A. Draw the radiation curve of blackbody B in the same graph.

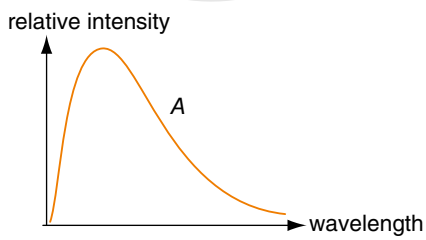


Fig a

★**6** A spherical object X is a blackbody of surface area 1.5 m^2. The power radiated per unit surface area of X is 3×10^3 W m^{-2}.

Take $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

(a) Find the luminosity and surface temperature of object X.

(b) Is object X a luminous or a non-luminous object? Explain briefly.

★**7** The surface temperatures of stars X and Y are 4000 K and 8000 K respectively. The size of Y is about the same as the Sun. The total power radiated by Y is P. The radius of X is 200 times that of Y.

(a) Find the total power radiated by X in terms of P.

(b) Which group of stars do X and Y most likely belongs to in the H-R diagram?

★**8** (a) A class K main sequence star is orange while a class B main sequence star is bluish-white. Which class of stars has a higher surface temperature?

(b) Planet Uranus (Fig b) is bluish and planet Venus (Fig c) is reddish. It is claimed that the surface temperature of Uranus is higher than that of Venus, as blue light has a shorter wavelength than red light. Explain whether you agree with this statement.



Fig b



Fig c

★**9** The figure below shows the radiation spectrum of the Sun. There are some dark lines in the spectrum (Fig d).



Fig d

(a) What are these dark lines called?

(b) Explain briefly how the radiation spectrum is formed.

(c) Suggest one piece of information about the Sun that can be obtained from the spectrum.

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3.4 The Doppler effect in astronomy

Have you got an experience when a fire truck passes you with its siren blaring (Fig 3.4a)?

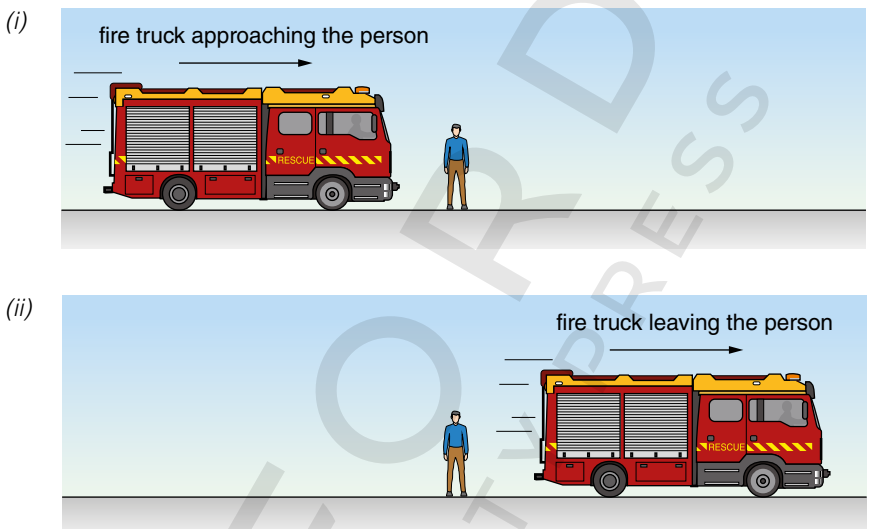


Fig 3.4a *A person hears (i) a higher pitch and (ii) a lower pitch from a fire truck siren.*

Simulation

→ This simulation allows Ss to experience the change in pitch of sound when Doppler effect occurs. It is available on OUP web in PC version (.exe) only.

The sound of siren has a higher pitch when the fire truck is approaching you than when it is leaving you. Why does it happen? How is it related to the evidence that suggests our universe is expanding?

DSE goal

Realize the Doppler effect and apply $\frac{\Delta\lambda}{\lambda_0} \approx \frac{v_r}{c}$ to determine the radial velocity of celestial bodies.

DSE exam

14(2)Q1.4, 15(2)Q1.8, 18(2)Q.1, 19(2)Q1.5, 20(2)Q1.5, Q.1(d), 21(2)Q1.7, Q1.8, 22(2)Q1.7, 23(2)Q.1(c)

1 The Doppler effect

a The Doppler effect of sound

Suppose a stationary fire truck gives out a siren sound of a constant frequency. The sound waves spread outwards evenly and the frequency of the sound you hear is the same everywhere (Fig 3.4b).

Flipped classroom

Watch a video and answer the questions.

Simulation

→ This simulation demonstrates the Doppler effect by showing the wavefronts. It is available on OUP web in PC version (.exe) only.



Fig 3.4b *The sound wave given out by a stationary source.*

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However, when the fire truck is moving forwards, the waves in front of it are effectively 'compressed' and the waves behind are 'stretched' (Fig 3.4c). Therefore, as the truck is approaching, you would hear a higher pitch, i.e. a shorter wavelength and a higher frequency. As the truck is leaving, you would hear a lower pitch, i.e. a longer wavelength and a lower frequency.

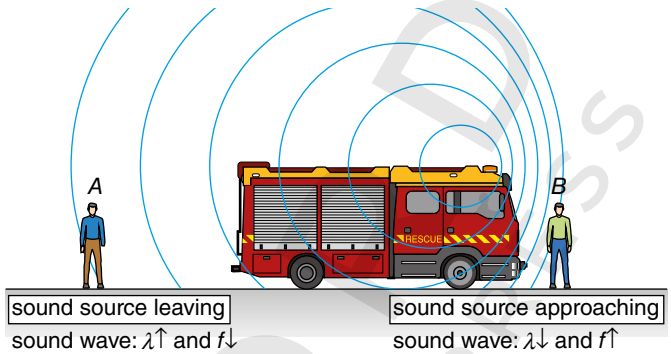


Fig 3.4c *Wavelength is changed along the direction of motion of the source.*

This phenomenon is called the **Doppler effect**. It is caused by the relative motion between the observer and the wave source. All kinds of waves exhibit the Doppler effect, including light waves.

Video

→ This video demonstrates the Phone experiment.

You may search the app on the Internet.

tone generator app

sound frequency analyser app

Phone experiment

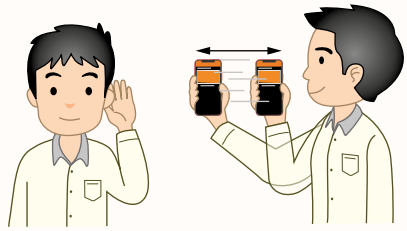
→ Worksheet is available on OUP web.

Demonstrating the Doppler effect for sound

What to do

1 Prepare two smartphones *X* and *Y*. Install a tone generator app in *X* and a sound frequency analyser app in *Y*.

2 Play a sound of 1000 Hz on *X*. Ask another person to move *X* towards and away from you quickly. Listen to the sound and pay attention to its pitch.

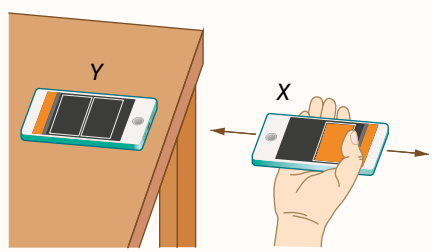


The volume of the sound should not be too loud.

The speaker should be kept at an appropriate distance with the ears of you and the person in help.

The speaker of *X* and the microphone of *Y* should face each other. The microphone and speaker of smartphone are usually located at the bottom.

3 Place *Y* on a table and hold *X* at the same height with *Y*. Play the tone again on *X*. Start measuring the frequency of sound on *Y*. Move *X* horizontally towards and away from *Y* quickly. Observe how the frequency measured by *Y* changes.



What you find

How is the frequency measured related to the motion of *X*?

Doppler effect 多普勒效應

b The Doppler effect of light

Similar to the sound wave, the Doppler effect of starlight is caused by the relative motion between the Earth and a star.

When a star is approaching us, all wavelengths of its light appear to be shorter. Visible light shifts towards the blue end of the spectrum. This change in colour is called a **blue shift**.

Conversely, the light from a star leaving us appear to have longer wavelengths. Visible light shifts towards the red end of the spectrum, i.e. **red shift** occurs.

Both the blue shift and the red shift are called **Doppler shift**.

Suppose a star emits radiation of wavelength λ_0. When it moves away from the Earth **along the line of sight**, the wavelength observed from the Earth is shifted to λ due to the Doppler effect (Fig 3.4d). The change in wavelength, $\Delta\lambda = \lambda - \lambda_0$, is related to the speed of the star by the equation:

The line of sight here refers to an imaginary straight line joining the Earth and the star.

The equation is valid only for stars moving at speeds much smaller than the speed of light.

$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda - \lambda_0}{\lambda_0} \approx \frac{\text{speed of star}}{\text{speed of light}} \quad \text{..... (1)}$$

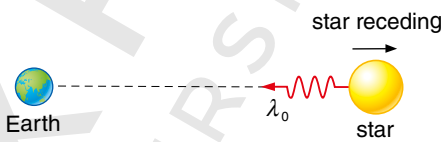


Fig 3.4d *The Doppler effect of a star moving along the line of sight.*

Generally, a star does not move along the line of sight. In these cases, we resolve the velocity of the star into two components (Fig 3.4e):

1 Transverse velocity—in direction perpendicular to the line of sight

2 **Radial velocity**—in direction along the line of sight

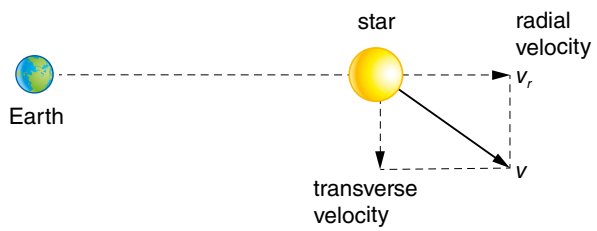


Fig 3.4e *Transverse and radial velocities of a star.*

Then the speed of the star in equation (1) corresponds to its radial velocity v_r. That is,

$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda - \lambda_0}{\lambda_0} \approx \frac{v_r}{c} \text{ (for } v_r \ll c\text{)} \quad \text{..... (2)}$$

speed of light

blue shift 藍移 red shift 紅移 Doppler shift 多普勒頻移 radial velocity 徑向速度

The sign convention of equation (2) for red and blue shifts is as follows (Table 3.4a):

	$\Delta\lambda$	v_r
Red shift (star is receding)	+ ($\because \lambda > \lambda_0$)	+
Blue shift (star is approaching)	− ($\because \lambda < \lambda_0$)	−

Table 3.4a *The sign convention for the equation of Doppler shift.*

By comparing the absorption lines in the spectrum of a star with the spectral lines of the same elements in a stationary source in the laboratory (Fig 3.4f), the **radial velocity** of the star can be found using equation (2).

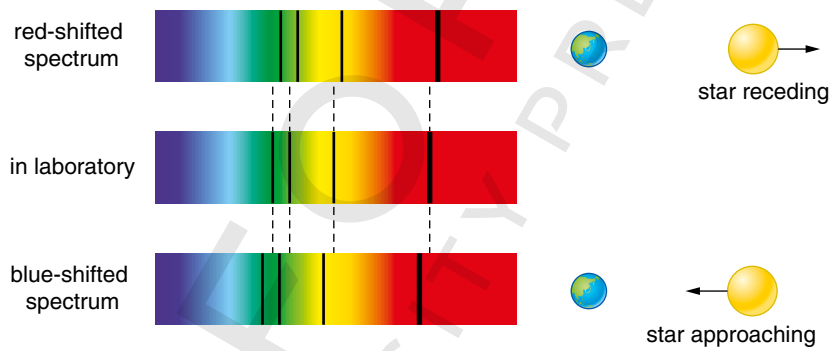


Fig 3.4f *Absorption spectra showing the relative motion between the Earth and a star.*

Know more

Christian Doppler (1803–1853)



Christian Doppler is an Austrian physicist. In 1842, he tried to explain the colour of light given out by double stars. Meanwhile, he proposed that the frequency of light from a source depends on its velocity relative to an observer. The effect for light was too small to measure at Doppler's time, but the effect for sound was soon observed.

Example 7 Doppler effect of a star

We observe a line in the spectrum of a star with a wavelength of 600.1 nm. Laboratory measurements show that the line should have an unshifted wavelength of 600.0 nm.

(a) Find the radial velocity of the star. Take $c = 3.00 \times 10^8$ m s^{-1}.

(b) Deduce and explain whether the star is approaching or receding from the Earth.

Solution

(a) By $\dfrac{\Delta\lambda}{\lambda_0} \approx \dfrac{v_r}{c}$,

$$\text{radial velocity} \approx \left(\frac{\Delta\lambda}{\lambda_0}\right)c = \frac{600.1 - 600.0}{600.0} \times 3.00 \times 10^8 = 5 \times 10^4 \text{ m s}^{-1}$$

The magnitude of the radial velocity is much greater than the speed of a space shuttle (at about 8000 m s^{-1} when coming back to the Earth). However, this speed is rather slow compared with other fast-moving stars.

(b) The positive sign for the radial velocity means that the star is moving away from the Earth.

Alternative solution:

The spectral line shifts to a longer wavelength. This indicates the star is moving away from the Earth.

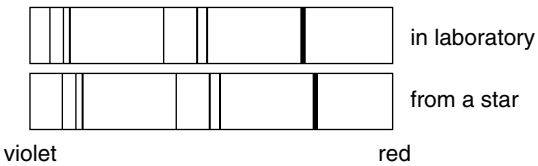
Christian Doppler 多普勒

Quick check 11

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The figure below shows spectra in laboratory and from a star.



Determine whether the following statements are true.

(a) The spectral lines from the star are blue-shifted. (T/F)

(b) The star is approaching the Earth. (T/F)

2 *True or false:*

(a) A star appearing red must be moving away from us. (T/F)

(b) A star moving towards us must be blue in colour. (T/F)

Learn by practice 6

1 The violet line (410 nm) of the hydrogen spectrum from a distant celestial body appears 5 nm shorter than the reference value. Take $c = 3.00 \times 10^8$ m s^{-1}.

(a) Describe the motion of the celestial body relative to the Earth.

(b) Find the radial velocity of the celestial body.

(c) Find the change in wavelength of the red line (656 nm) from the same source.

2 Applications of the Doppler effect

DSE goal

Use the radial velocity curve to determine the orbital radius, speed, and period of a small celestial body in circular orbital motion around a massive body as seen along the orbital plane.

DSE exam

15(2)Q.1, 21(2)Q.1(b)

a In search of exoplanets

Planets outside our solar system are known as *exoplanets*. Since they do not emit light and are easily overshadowed by the parent star, they can hardly be seen directly. One way to find them is to use the Doppler effect.

As a planet orbits a star under their mutual gravitational pull, both of them revolve around their common centre, the **centre of mass**, with the same orbital period (Fig 3.4g). However, since the star always has a larger mass than its planet, it moves in a much smaller orbit than the planet. ▶

This is true for our solar system. Since the Sun is much more massive than all of its planets, the centre of mass of the whole system still lies inside the Sun. The Sun's orbit is so small that we assume it to be stationary when discussing planetary motion in the previous chapters.

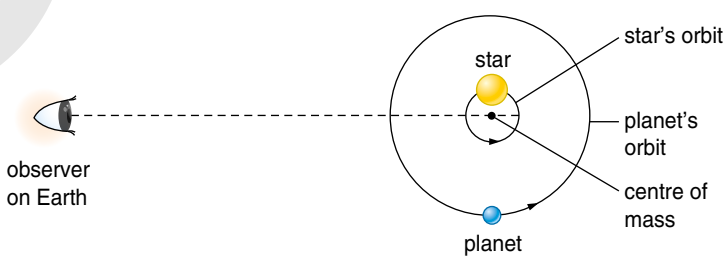


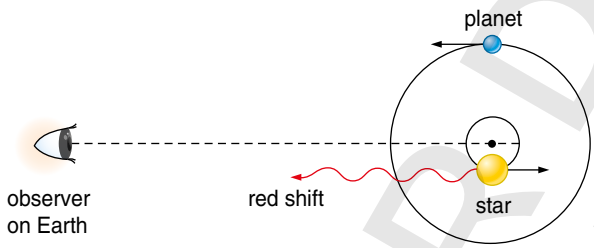
Fig 3.4g *The extrasolar planet and its star orbit around their centre of mass.*

exoplanet 太陽系外行星

Therefore, if a star shows periodic Doppler shifts, it is likely to have another celestial body orbiting around it. The celestial body could be a planet.

▶ The orbital motion of the star (Fig 3.4h) results in periodic blue shift and red shift in its spectrum. From the spectrum of the star, the radial velocity of the star at different instants can be calculated by $v_r \approx \left(\frac{\Delta\lambda}{\lambda_0}\right) c$.

(i) The star is moving away from the observer.



(ii) The star is moving towards the observer.

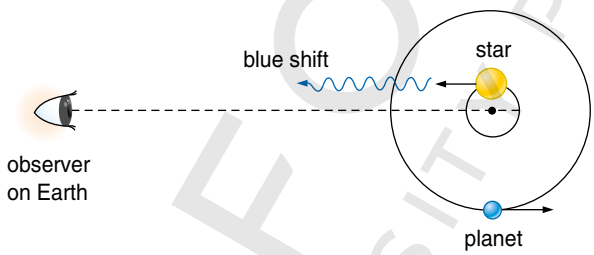
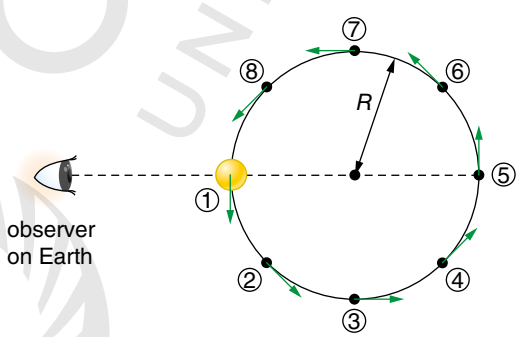


Fig 3.4h *Periodic blue and red shifts can be detected from a star with planet.*

By plotting radial velocity v_r against time t, we obtain a **radial velocity curve** (Fig 3.4i).

(i)



(ii)

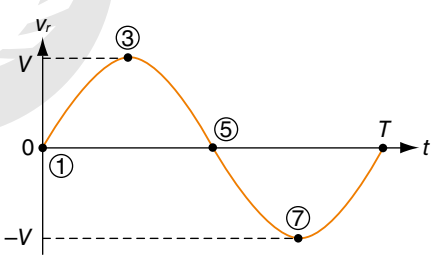


Fig 3.4i *(i) The motion of the star at different positions in its orbit. (ii) The radial velocity curve of a star with a planet.*

radial velocity curve 徑向速度曲線

For simplicity, we assume that the orbit is circular and the Earth lies on the same orbital plane.

▶ The radial velocity curve gives us the following information:

1 **The orbital period T of the planet and the star**

2 **The angular velocity ω of the planet and the star**

As the planet and the star have the same T, they have the same angular velocity ω.

$$\omega = \frac{2\pi}{T}$$

3 **The orbital speed V of the star—the maximum radial velocity**

4 **The orbital radius R of the star**

$$\text{By } V = \frac{2\pi R}{T} = Rw,$$

$$R = \frac{VT}{2\pi} = \frac{V}{\omega}$$

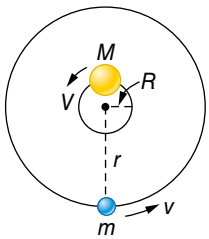


Fig 3.4j *Circular motion of a star and its planet.*

Consider the circular motion of the planet (Fig 3.4j). The gravitational pull between the star and the planet also provides the centripetal force for the planet.

$$F = \frac{GMm}{(R+r)^2} = \frac{mv^2}{r} = mr\omega^2$$

where M and m are the masses of the star and the planet respectively, v is the orbital speed of the planet and r is the orbital radius of the planet.

As mentioned, a star is generally much more massive than its planet. Their centre of mass will be very close to the centre of the star. In this case, $R \ll r$ and we assume R to be negligible. The above equation then becomes

$$F = \frac{GMm}{r^2} = mr\omega^2$$

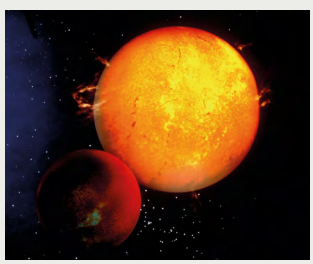
Astronomers can estimate the mass M of a star from other observation data. Therefore, r can be found.

▶ $$\Rightarrow r = \sqrt[3]{\frac{GM}{\omega^2}}$$

Know more

First exoplanet found around a main-sequence star

In 1995, astronomers *Michel Mayor* and *Didier Queloz* found the first exoplanet orbiting a solar-type star. The planet is named 51 Pegasi b, orbiting the star 51 Pegasi. It is about 50 ly away from the Earth, in the constellation of Pegasus. For this discovery, the two astronomers were awarded the 2019 Nobel Prize in Physics.

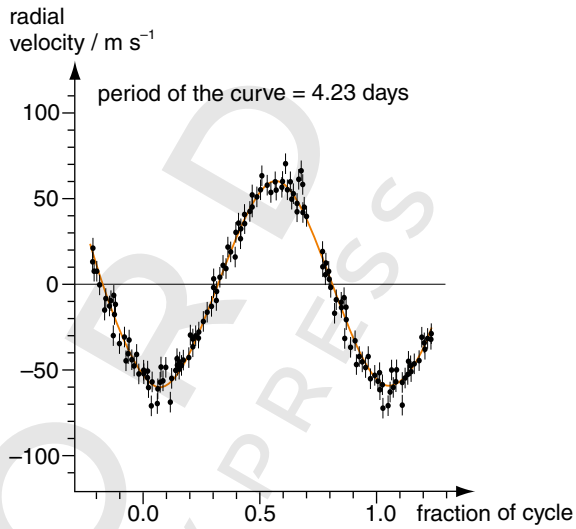


An artist's impression of 51 Pegasi b (in the foreground).

Michel Mayor 馬約爾 Didier Queloz 奎洛茲

Example 8 The radial velocity curve of 51 Pegasi

A planet is discovered to orbit the star 51 Pegasi. By measuring the Doppler shift, the radial velocity of 51 Pegasi as observed on Earth is deduced. The graph shows the change in radial velocity against time. It is known that the mass of 51 Pegasi is 2.10×10^{30} kg. Assume that we are looking along the orbital plane of 51 Pegasi and its planet, and their orbits are circular.



(a) Find the orbital periods and angular velocities of 51 Pegasi and its planet.

(b) Find the orbital speed and orbital radius of 51 Pegasi.

(c) Calculate the orbital radius and hence the orbital speed of the planet. Assume that the mass of 51 Pegasi is much larger than that of its planet. Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

Solution

(a) Period of the planet = period of 51 Pegasi
= period of the radial velocity curve
= 4.23 days

51 Pegasi and its planet have the same angular velocity ω.

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{4.23 \times 24 \times 3600} = 1.719 \times 10^{-5} \text{ rad s}^{-1} \approx 1.72 \times 10^{-5} \text{ rad s}^{-1}$$

(b) Orbital speed of 51 Pegasi = amplitude of the radial velocity curve = 60 m s^{-1}

Let R be the orbital radius of 51 Pegasi.

By $v = R\omega$,

$$R = \frac{v}{\omega} = \frac{60}{1.719 \times 10^{-5}} = 3.49 \times 10^6 \text{ m}$$

(c) Let M and m be the masses of 51 Pegasi and its planet, r be the orbital radius of the planet. Assume R to be negligible as the mass of 51 Pegasi is much larger than that of its planet. Consider the gravitational pull between the star and the planet.

By $\frac{GMm}{r^2} = mr\omega^2$,

$$r = \sqrt[3]{\frac{GM}{\omega^2}} = \sqrt[3]{\frac{(6.67 \times 10^{-11})(2.10 \times 10^{30})}{(1.719 \times 10^{-5})^2}} = 7.79 \times 10^9 \text{ m}$$

$\therefore$ Orbital speed of the planet = $r\omega = (7.79 \times 10^9) \times (1.719 \times 10^{-5}) = 1.34 \times 10^5$ m s^{-1}

Exercise 3.4 Q8 (p.141)

b In search of binary systems

Some star systems may consist of even more stars.

Our Sun exists as a single star is in the minority in the universe. In fact, most stars exist in binary star systems or systems with three or more stars.

A binary system is a system that consists of two stars orbiting around their centre of mass. It can be detected by analysing the Doppler shifts of spectral lines of stars. Like the exoplanet and its star, we can obtain information of the stars in a binary system using the Doppler effect and their radial velocity curves.

Example 9 Stars in a binary system

Stars X and Y are suspected to be stars in a binary system. The radial velocity curves for the two stars are shown in Figure a. The maximum shift in wavelength of the spectral lines from X is shown in Figure b.

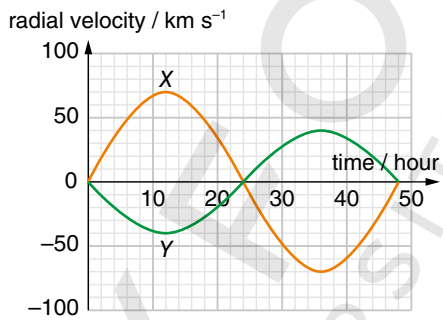


Fig a

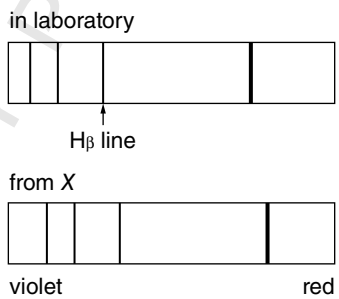


Fig b

(a) State the evidence to support that X and Y are stars in a binary system.

(b) Refer to the radial velocity curve of X. Which moment does the maximum shift in the spectral lines shown in Figure b correspond to?

(c) The wavelength of the H_β line obtained in a laboratory is 486.4 nm. What is the maximum Doppler shift of the line from X? Take $c = 3.00 \times 10^8$ m s^{-1}.

Solution

(a) X and Y revolve with the same period in antiphase.

(b) At time = 12 hours.

(c) Maximum radial velocity of X = amplitude of curve X = 70 km s^{-1}

$$\text{By } \frac{\Delta\lambda}{\lambda_0} = \frac{v_r}{c},\ \Delta\lambda = \left(\frac{v_r}{c}\right) \times \lambda_0 = \frac{70 \times 10^3}{3.00 \times 10^8} \times 486.4 = 0.113 \text{ nm}$$

The maximum Doppler shift of the H_β line observed is 0.113 nm.

Caution
Unlike an exoplanetary system, the stars in a binary system may be of comparable mass. The orbital radius of both stars must be considered in the calculation.

Think about...

Assume that the orbits of the stars are circular and the observer lies on the same orbital plane. Deduce the orbital radii and hence the masses of X and Y. Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

Google Forms Microsoft Forms

Quick check 12

→ E-version in OUP Exercise Platform is also available.

1 Consider two stars in a binary system. Assume that their orbits are circular and they are observed along their orbital plane. Determine whether the following statements are true.

(a) The two stars must have the same orbital period. (T/F)

(b) Each star revolves around different and separate centres. (T/F)

(c) The two stars must have the same size of angular velocity. (T/F)

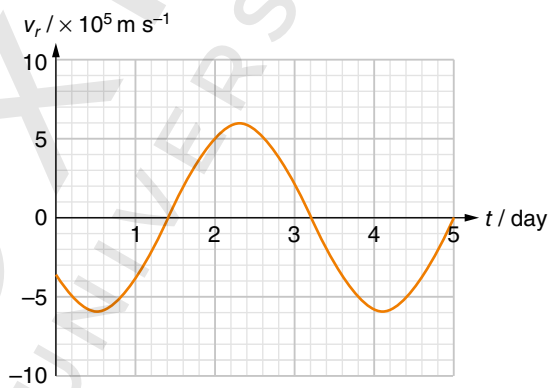
(d) The mass of each star can be directly read from their radial velocity curves. (T/F)

(e) The orbital speeds of the two stars can be found from their radial velocity curves. (T/F)

Learn by practice 7

Take $c = 3.00 \times 10^8$ m s^{-1}, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

1 A planet is found to orbit a star in a circular path. The figure below shows the radial velocity v_r of the star plotted against time t.



Assume that the Earth lies on the orbital plane of the star and its planet.

(a) At which time interval does the star move away from the observer?

(b) Find the orbital speed of the star.

(c) Find the orbital period of the star.

(d) Calculate the orbital radius of the star.

2 The edge-on view of a binary system shows that the wavelength of the calcium H-line of one of the stars changes by ±0.3 nm from the same line (λ = 396.9 nm) obtained in a laboratory. The period of the star is 72 hours.

(a) What is the orbital speed of the star in km s^{-1}?

(b) Find the orbital radius of the star.

(c) Suppose the two stars in the binary system have equal mass and orbital radius. Estimate the mass of each star.

☆ HKDSE 2015 Paper 2 Q.1

→ This video demonstrates how this question is solved.

Video

DSE practice 3 | Binary system

An observer on Earth finds a binary system with star P more massive than star Q (Fig a). They move in circular orbits about the same centre of mass. Assume that all observations are made along their orbital plane. Figure b shows the radial velocity curves of the two stars. The period for one complete cycle is 0.78 day.

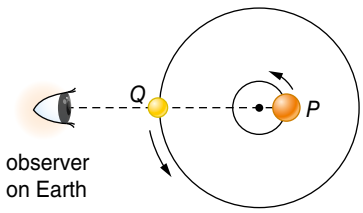


Fig a

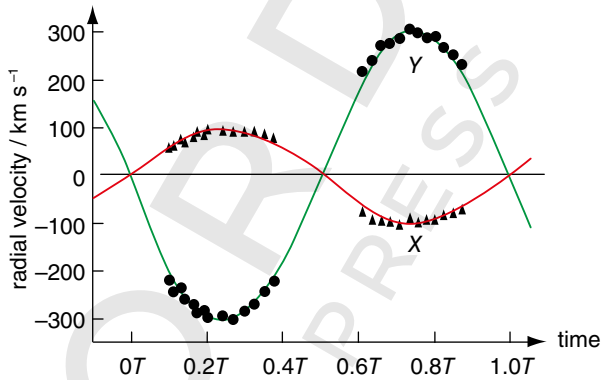


Fig b

(a) Which curve, X or Y, corresponds to star P? (1 mark)

(b) Refer to the radial velocity curves. Which instant does Figure a correspond to? (1 mark)

(c) Find the orbital speed v_P and orbital radius r_P of P. Using a similar method, find the orbital radius r_Q of Q. (4 marks)

(d) Calculate the mass m_P of P. Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}. (2 marks)

Solution

(a) Curve X. 1A

(b) Time = $0.5T$ 1A

(c) v_P = amplitude of curve X = 100 km s^{-1} 1A

$$v_P = \frac{2\pi r_P}{T} \quad \text{1M}$$

$$r_P = \frac{Tv_P}{2\pi} = \frac{(0.78 \times 24 \times 3600)(100 \times 10^3)}{2\pi}$$

$$= 1.073 \times 10^9 \text{ m} \approx 1.07 \times 10^9 \text{ m} \quad \text{1A}$$

Similarly, $v_Q = 300$ km s^{-1}.

$$r_Q = \frac{Tv_Q}{2\pi} = \frac{(0.78 \times 24 \times 3600)(300 \times 10^3)}{2\pi}$$

$$= 3.218 \times 10^9 \text{ m} \approx 3.22 \times 10^9 \text{ m} \quad \text{1A}$$

(d) The gravitational pull provides the centripetal force.

$$\frac{Gm_Pm_Q}{(r_P + r_Q)^2} = \frac{m_Qv_Q^2}{r_Q} \quad \text{1M}$$

$$m_P = \frac{v_Q^2(r_P + r_Q)^2}{Gr_Q} = \frac{(300 \times 10^3)^2(1.073 \times 10^9 + 3.218 \times 10^9)^2}{(6.67 \times 10^{-11})(3.218 \times 10^9)}$$

$$= 7.72 \times 10^{30} \text{ kg} \quad \text{1A}$$

Since P and Q have equal period and P travels a shorter distance than Q in one cycle, P must have a lower orbital speed.

Common mistake ⚠
Students may wrongly think that the gravitational force is $\frac{Gm_Pm_Q}{r_Q^2}$. In binary system, the masses of both stars may be comparable. The one with smaller orbit cannot be considered as stationary.

▶ Revision 3 Q33 (p.153)

DSE goal

Relate the rotation curve of stars around galaxies to the existence of dark matter.

DSE exam

13(2)Q1.8, 18(2)Q1.8, 19(2)Q1.8

C Provide evidence for the existence of dark matter

In a galaxy, all matter, including stars, orbits around the galactic centre. The Doppler effect can be used to measure the stars' radial velocities, which in turn help deduce their orbital speeds. When their orbital speeds are plotted against their distances from the galactic centre, we obtain the **rotation curve** of the galaxy (Fig 3.4k).

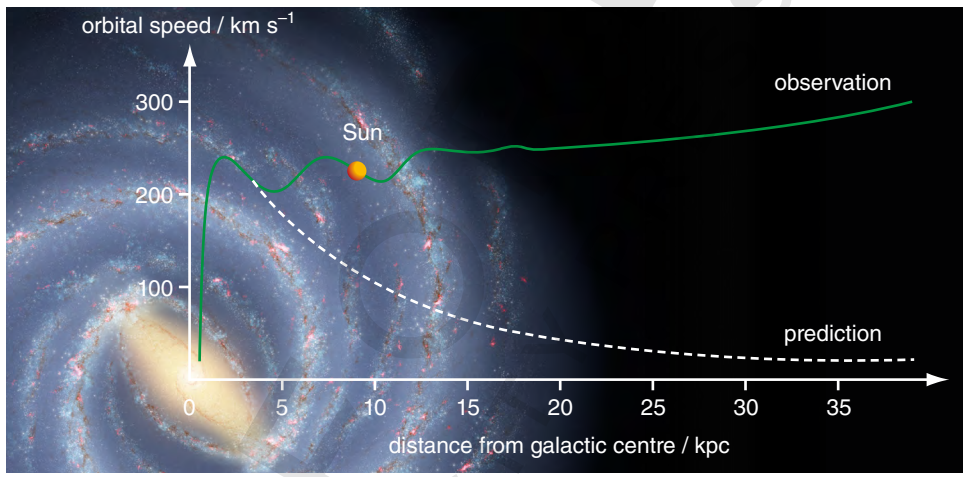


Fig 3.4k *Rotation curve of Milky Way Galaxy.*

Most of the luminous mass of a galaxy is concentrated at its centre, just like a star-planet system. If the luminous mass were all the matter existed, the orbital speed v should decrease when the distance r from the galactic centre increases, as the dotted line shown in Figure 3.4k.

If all the mass of a galaxy is concentrated at its centre, $v \propto \frac{1}{\sqrt{r}}$.

However, observation shows that further stars revolve around the galactic centre at equal or increasing speeds. This contradicts the above prediction and suggests that there should be much more matter than we have actually observed at a large distance from the galactic centre. This missing matter is called **dark matter**. The gravitational pull of dark matter provides the centripetal force to keep the stars at high orbital speed.

Dark matter does not necessarily refer to black holes.

Dark matter does not emit or reflect any electromagnetic radiation. It is invisible to us and has not been detected directly with current instruments. The nature of dark matter is not yet known, but it is certain that the total mass of dark matter in the universe is greater than that of all known matter.

Know more

Components of the universe

Apart from the known matter and dark matter, the universe is more complex than we imagine. Do you know at what percentage of the mass of the universe has been explored?

▸ p.156

rotation curve 旋轉曲線　　dark matter 暗物質

DSE goal

Relate the red shift to the expansion of the universe.

DSE exam

16(2)Q1.8, 21(2)Q1.6, 22(2)Q.1(c)

d Provide evidence for the expansion of the universe

Astronomers have found that the spectra of very distant galaxies are red-shifted. *Edwin Hubble* interpreted this as these galaxies are moving away from our galaxy. He summarized the relationship between the receding velocity and the distance to these galaxies in Hubble's law (Fig 3.4l):

The greater the distance d from the Earth, the higher the receding velocity v of a galaxy, i.e. $d\uparrow \Rightarrow v\uparrow$.

Hubble's law only works for distant galaxies (d > 10 Mpc). It is usually expressed mathematically in the form $v = Hd$ where H is the Hubble constant.

The Hubble constant characterizes the present-day expansion rate of the universe. Its value is still controversial, but it is generally agreed that it should be around 70 km s^{-1} Mpc^{-1}.

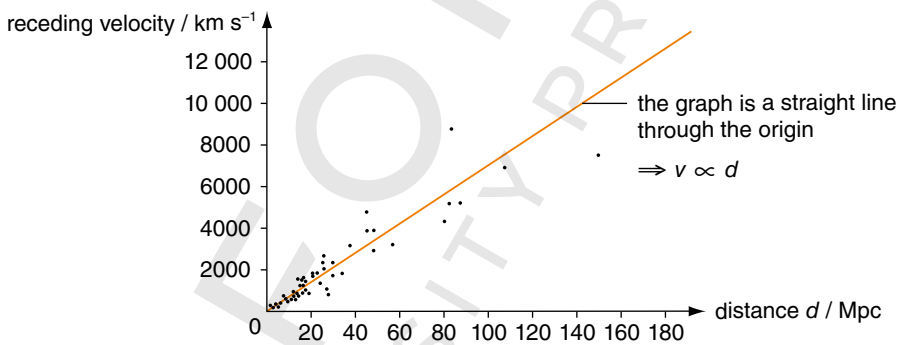


Fig 3.4l *The receding velocity of a galaxy is directly proportional to its distance from the Earth.*

The distance is in Mpc. The size of a galaxy cluster is about 1–5 Mpc.

The red shift due to expansion is obvious only for distant galaxies. Bodies within the Local Group are bound by gravity and the effect of the expansion cannot be seen.

Hubble's law can be explained if the universe (or space itself) is expanding. The expansion is similar to the baking of bread (Fig 3.4m). As the *dough* rises (the size of the bread increases), a *raisin* would 'see' that all the other raisins in the bread are moving away. Raisins that were originally close together would move apart more slowly, and raisins originally further apart would move away from each other more rapidly.

Unlike the bread, the universe has neither a boundary nor a centre.

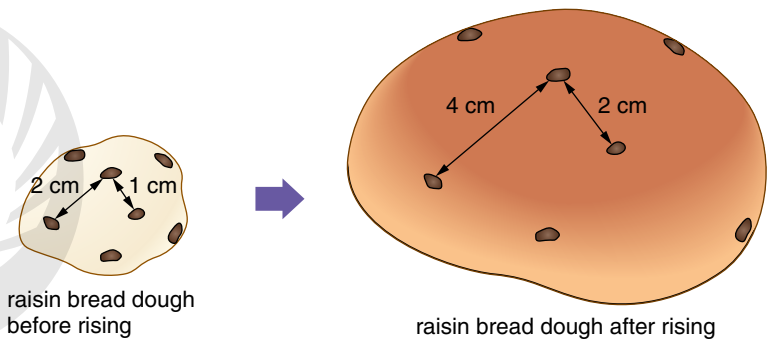


Fig 3.4m *Expansion of the universe is similar to the baking of bread.*

The universe is ever expanding. At a certain point back in time, everything in the universe would have been extremely close together. From that point in time onwards, space has expanded at a high speed. This expansion is known as the **Big Bang**.

Know more

Edwin Hubble (1889–1953)



Edwin Hubble was an American astronomer. He discovered the existence of other galaxies besides the Milky Way and the expansion of the universe. The well-known Hubble Space Telescope is named after him.

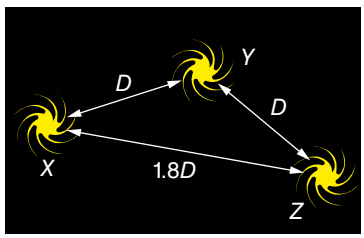
Edwin Hubble 哈勃　dough 麪糰　raisin 提子乾　Big Bang 大爆炸

Google Forms Microsoft Forms

Quick check 13

→ E-version in OUP Exercise Platform is also available.

1 Consider galaxies *X*, *Y* and *Z* below. They are more than 10 Mpc apart from each other.



Determine whether the following statements are true.

(a) Light from both *X* and *Z* show roughly the same degree of red shift when observed from *Y*. (T/F)

(b) The Doppler shift of spectral lines of *Y* is greater than that of *Z* when observed from *X*. (T/F)

(c) *X* is moving away at a higher speed than *Y* is when observed from *Z*. (T/F)

2 Which of the following statements about dark matter is correct?

A It can be detected directly using ultrasound.

B It does not affect the motion of stars.

C It exists only at the galactic centre.

D Its existence is deduced by studying the rotation curves of galaxies.

3 Which of the following statements is correct?

A The spectra of very distant galaxies are blue-shifted.

B The expansion of the universe refers to the fact that most galaxies in the universe are expanding.

C Everything in the universe was closer together in the past.

D The Big Bang refers to the explosion of galaxies.

Exercise 3.4

Take $c = 3.00 \times 10^8$ m s^{-1}, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

1 A planet is found to be orbiting star *X*. Which of the following statements about the radial velocity curve of star *X* is/are correct?

(1) The orbital radius of star *X* can be directly read from the curve.

(2) It is obtained by studying the Doppler effect of star *X*.

(3) It represents periodic red and blue shifts.

A (1) only

B (1) and (2) only

C (1) and (3) only

D (2) and (3) only

2 A star is found to move around a stationary star in a circular orbit (Fig a). The radial velocity curve of the orbiting star in Figure b shows four moments *P*, *Q*, *R* and *S*. Assume that the orbital plane of the star is along the line of sight.

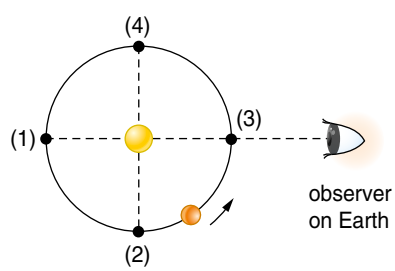


Fig a

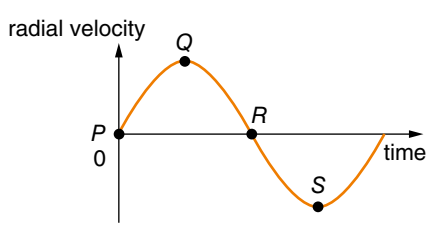


Fig b

Which of the following is the correct combination of positions in the orbit and moments on the curve?

	(1)	(2)	(3)	(4)
A	*P*	*Q*	*R*	*S*
B	*Q*	*R*	*S*	*P*
C	*R*	*S*	*P*	*Q*
D	*S*	*P*	*Q*	*R*

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3 Which of the following statements is correct?

A Starlight from a star is red-shifted if the star is moving towards us.

B The Doppler effect for starlight depends on the relative motion between the star and the observer.

C Starlight shifts to become blue in colour when blue shift occurs.

D A larger Doppler shift of spectral lines of a star means the star has a lower radial velocity.

★**4** Which of the following statements is/are correct?

(1) The stars in the Milky Way Galaxy orbit around the galactic centre.

(2) The rotational velocities in the outer part of the galaxy are smaller than expected.

(3) The existence of dark matter affects the motion of stars in a galaxy.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**5** Galaxies *P*, *Q* and *R* are separated by equal distance *x* (Fig c).

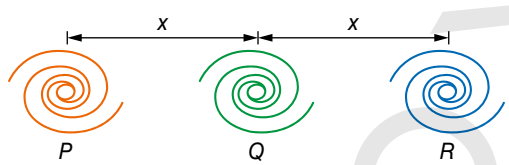


Fig c

After a period of time, the distance between *P* and *Q* is doubled. What is the ratio of the velocity of *Q* to that of *R* as observed from *P*?

A 1 : 1 B 1 : 2

C 2 : 1 D 1 : 4

6 A star is approaching the Earth at a radial velocity of 25 km s^{-1}. At what wavelength do we see a spectral line that is found to be 500.00 nm when unshifted?

7 A spectral line at 659.5 nm is found in the spectrum of a distant star. Its wavelength appears 3.2 nm longer than that observed in a laboratory. What is the wavelength observed in the laboratory if there is another spectral line at 488.5 nm in the spectrum of the star?

8 Suppose the spectral lines of the following stars can be detected on Earth. Complete the table below.

	Star	λ / nm	λ_0 / nm	v_r / km s^{-1}	Blue shift / red shift?
(a)	*X*	452.25	451.86		
(b)	*Y*		640.05	–534	
(c)	*Z*	583.76	584.13		

9 Galaxies *X* and *Y* are both more than 100 Mpc away from the Earth. The respective absorption spectra of hydrogen from *X*, in the laboratory, and *Y* are shown in Figure d. Which galaxy, *X* or *Y*, is further away from the Earth? Explain.

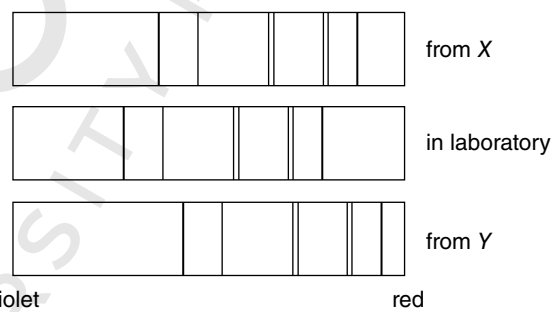


Fig d

★**10** Four stars *A*, *B*, *C* and *D* are observed on Earth as shown (Fig e). All the stars are moving at the same speed. The arrows show the moving direction of the stars.

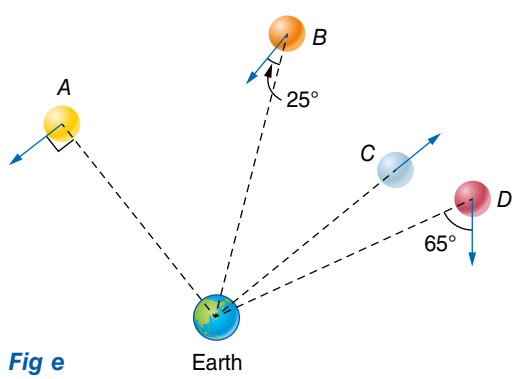


Fig e

(a) Which star(s) show(s) a red shift?

(b) Which star(s) do(es) not show any Doppler effect on the spectrum?

(c) Except for the star(s) in the answer of (b), which star has the smallest radial velocity as observed from the Earth?

★**11** Star P orbits star Q, which is much more massive than star P. Assume that there is an observer at the centre of the orbit of P. Explain how the observer can verify that the orbit of star P is circular by studying its spectrum.

★**12** A space traveller is approaching the solar system at a speed of 100 km s^{-1} relative to the Sun (Fig f) and is observing a 650-nm red light beam coming from the Earth. The traveller's trajectory lies on the same plane as the Earth's orbit. Take the radius of Earth's orbit to be 1.50×10^{11} m. Take 1 yr = 3.16×10^{7} s.

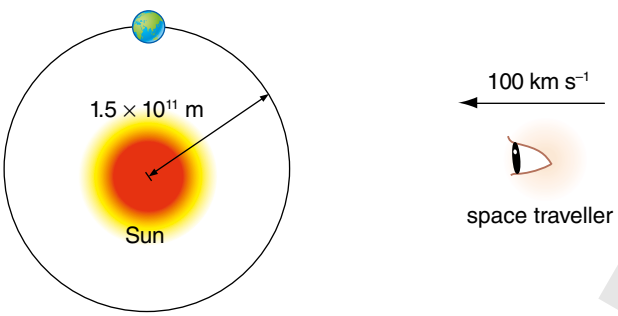


Fig f

(a) Find the minimum and maximum speeds of the Earth as seen by the space traveller.

(b) Hence, find the minimum and maximum wavelengths of the red light that the traveller observes as the Earth orbits the Sun.

★**13** The edge-on view of a spiral galaxy can be observed on Earth as shown below (Fig g). Stars X and Y are located on the opposite side of the galactic centre. The largest angular separation between them is 1.8° as observed from the Earth. The galaxy is 560 kpc away from the Earth.

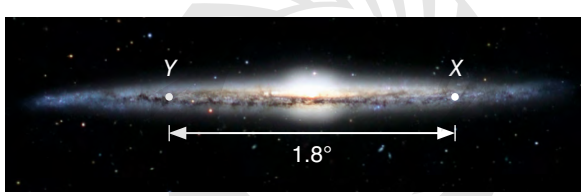


Fig g

(a) Estimate the distance (in m) between X and Y. Take 1 pc = 3.09×10^{16} m.

(b) The wavelengths of a spectral line from X and Y are measured to be 486.63 nm and 485.54 nm respectively. The wavelength of the same spectral line from a stationary source on Earth is measured to be 486.01 nm. Estimate the radial velocity of X and Y.

(c) X and Y are equidistant from the galactic centre. The magnitudes of their radial velocities should be about the same. The discrepancy shown in the answers to (b) is caused by the motion of the galaxy. Is the galaxy moving closer to or further away from the Earth? Explain briefly.

(d) The rotation speed of matter at radius R from the galactic centre is V. The radius of the galactic disc is r. The rotation curve of the galaxy is shown below (Fig h).

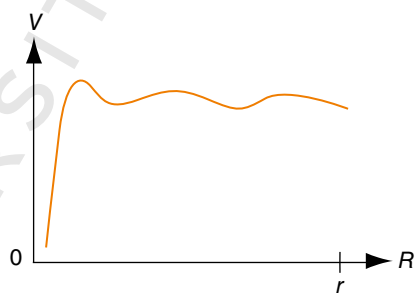


Fig h

The rotation curve suggests the existence of dark matter. What can you infer about the distribution of the dark matter from the shape of the curve?

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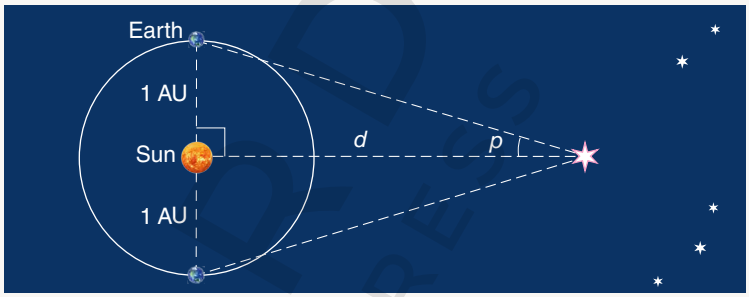
Key concepts 3

1 Method of parallax

- Used to measure the distance of nearby stars

Parsec (pc)

1 pc = distance of a star whose parallax is one arcsec
= 206 265 AU



$$d \text{ (in parsecs)} = \frac{1}{p \text{ (in arcsecs)}}$$

2 Brightness of star

From the observer's point of view (depend on distance d)		From the star's point of view (independent of distance)
Apparent brightness (b) Energy received by an observer per unit area per unit time $b = \frac{\text{total energy}}{\text{area} \times \text{time}} = \frac{L}{4\pi d^2}$	↔	**Luminosity (L)** Total energy given out by a star in form of EM radiation per unit time $L = \frac{\text{total energy emitted}}{\text{time}}$
↓ convenient scale for astronomy		↓ convenient scale for astronomy
Apparent magnitude (m) Measures the brightness of celestial bodies as seen by an observer on Earth. smaller m ⇒ brighter	↔	**Absolute magnitude (M)** The apparent magnitude of a star if it were 10 pc away smaller M ⇒ higher L

By comparing m and M, the distance of star may be estimated.

3

Classification of star

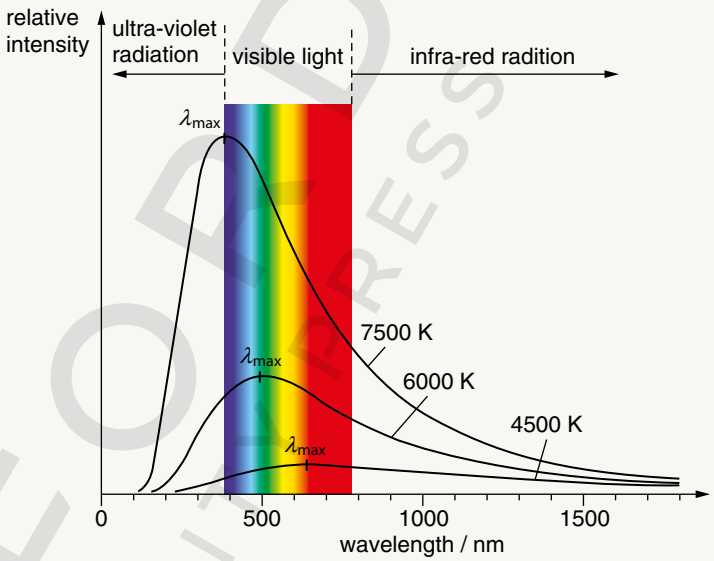
Stars behave like blackbodies which absorb all EM radiation falling on it and emit EM radiation at all λ.

Blackbody radiation curves

- Continuous at all λ
- Hotter star $\Rightarrow$ higher peak
- Curves do not intersect with each other
- Hotter star $\Rightarrow \lambda_{max} \downarrow$
  $\Rightarrow$ appears bluer
  Cooler star $\Rightarrow \lambda_{max} \uparrow$
  $\Rightarrow$ appears redder
- Area under curve
  = total power radiated per unit surface area (J)

Stefan's law: $J = \sigma T^4$

$$L = 4\pi R^2 \sigma T^4$$



4

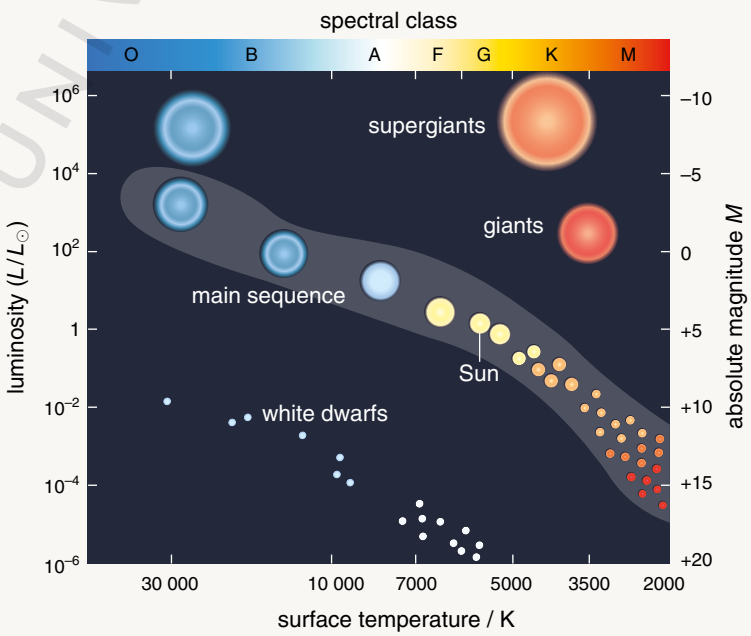
Spectral classes of stars

7 classes according to surface temperature: **O, B, A, F, G, K, M**

Hertzsprung-Russell diagram (H-R diagram)

3 main groups:

- main sequence
- giant / red giant
- white dwarf



5

Absorption spectrum



Spectral lines are due to the decrease in radiation intensity at certain wavelengths.

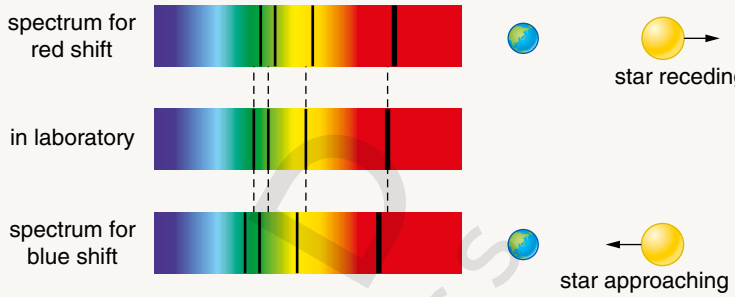
6 Doppler effect

- The change in the observed wavelength from a wave source due to the **relative motion** between the observer and the wave source

$$\frac{\Delta\lambda}{\lambda_0} \approx \frac{v_r}{c} \quad (\text{for } v_r \ll c)$$

where v_r = radial velocity

- Star is receding:
  $\lambda \uparrow \Rightarrow$ red shift
- Star is approaching:
  $\lambda \downarrow \Rightarrow$ blue shift



Applications of Doppler effect

- Finding extrasolar planet
- Finding binary stars
- Providing evidence for the existence of dark matter
- Providing evidence for the expansion of the universe



Key terms

1 absolute magnitude 絕對星等 p.108
2 absorption line 吸收譜線 p.120
3 absorption spectrum (pl. spectra) 吸收光譜 p.120
4 angle of stellar parallax 恆星視差角 p.99
5 angular diameter 角直徑 p.97
6 angular distance 角距離 p.97
7 angular size 角大小 p.97
8 apparent brightness 視亮度 p.105
9 apparent magnitude 視星等 p.107
10 Big Bang 大爆炸 p.139
11 blackbody 黑體 p.114
12 blackbody radiation 黑體輻射 p.114
13 blue shift 藍移 p.129
14 dark matter 暗物質 p.138
15 Doppler effect 多普勒效應 p.128
16 Doppler shift 多普勒頻移 p.129
17 giant 巨星 p.123
18 Hertzsprung-Russell diagram (H-R diagram) 赫羅圖 p.122
19 luminosity 光度 p.104
20 main sequence star 主序星 p.123
21 main sequence 主序 p.123
22 minute of arc (arcmin or ′) 角分 p.97
23 parallax 視差 p.98
24 radial velocity 徑向速度 p.129
25 radial velocity curve 徑向速度曲線 p.132
26 red giant 紅巨星 p.123
27 red shift 紅移 p.129
28 rotation curve 旋轉曲線 p.138
29 second of arc (arcsec or ″) 角秒 p.97
30 spectral class 光譜型 p.121
31 spectral line 光譜線 p.120
32 Stefan's law 斯特藩定律 p.117
33 supergiant 超巨星 p.123
34 white dwarf 白矮星 p.123

Revision 3

Unless otherwise specified, use the following data in the questions.

1 pc = 3.09×10^{16} m = 3.26 ly = 206 265 AU

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

$c = 3.00 \times 10^{8}$ m s^{-1}

$G = 6.67 \times 10^{-11}$ N m^{2} kg^{-2}

Multiple-choice questions

1 3.1 Which of the following statements must be correct?

(1) The parallax of a nearby star is larger than that of a more distant star.

(2) The larger the parallax, the larger the angular shift of a star.

(3) If the distance to a star is 1 pc, the maximum angular shift of a star over six months is 1″.

A (1) only B (3) only

C (1) and (2) only D (1) and (3) only

2 3.3 According to Stefan's law, what is the power radiated (in solar unit $L_\odot$) by a star with double the surface temperature of the Sun and one-tenth the radius of the Sun?

A $\frac{4}{25}L_\odot$ B $4L_\odot$

C $16L_\odot$ D $1600L_\odot$

3 3.4 Which of the following statements about the expansion of the universe is/are correct?

(1) Most galaxies are receding from the Earth.

(2) The receding velocity of a distant galaxy is proportional to its distance.

(3) Red shift of light from galaxies can be detected.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

4 Misc What happens to the blackbody radiation curve of a star if it is moving towards the Earth?

A It has no change at all.

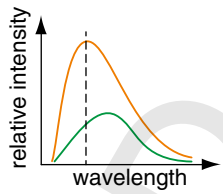
B It peaks at a lower intensity.

C It peaks at a longer wavelength.

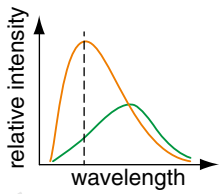
D It peaks at a shorter wavelength.

5 3.3 Which of the following sets of curves show(s) the blackbody radiation curves of the same blackbody at different surface temperatures?

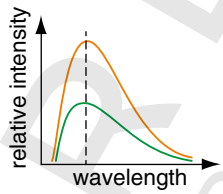
A



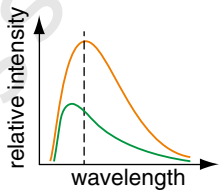
B



C



D



***6** 3.1 As observed from the Earth, the positions of Venus, Jupiter and the Moon on a particular day are shown in Figure a. They almost form an equilateral triangle in the sky.

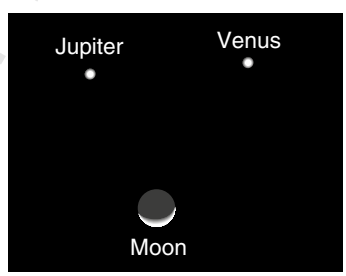


Fig a

Which of the following statements about the three celestial bodies must be correct?

(1) The angular distances between any two of them are roughly equal.

(2) The distances between any two of them are roughly equal.

(3) Their angular diameters are roughly equal.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

***7** 3.3 Which of the following statements is/are correct?

(1) A blackbody at 2380 K emits roughly twice as much radiation per unit surface area as the same blackbody at 2000 K.

(2) The power radiated per unit surface area of a blackbody is determined solely by its surface temperature.

(3) The surface temperature of a blackbody determines its colour.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

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(For Q8–9.) Star X is 2 pc away from the Earth. Figure b shows the relative positions of the Earth, the Sun and star X in February.

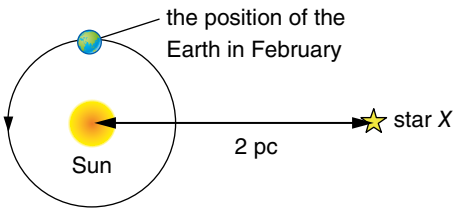


Fig b

Figure c shows the sky as observed from the Earth. Positions P, Q, R, S and T are 0.5″ apart from each other. Star X is at position R when the Earth is closest to it.



Fig c

★**8** 3.1 Which position in Figure c is the most possible position of star X in February?

A P B Q
C S D T

★**9** 3.1 Which position in Figure c is the most possible position of star X in August?

A P B Q
C S D T

★**10** 3.3 In Figure d, the luminosity and surface temperature of four stars, W, X, Y and Z, are measured with respect to the Sun. Which star(s) is/are larger than the Sun?

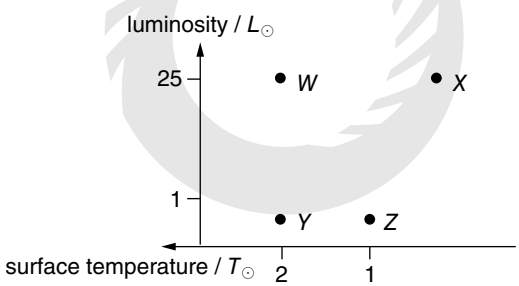


Fig d

A X only B W and X only
C W and Z only D Y and Z only

★**11** 3.3 Which of the following statements about blackbody radiation is/are true?

(1) A blackbody at higher temperature has higher intensity per unit surface area at all wavelengths than a cooler one.

(2) The position of the peak of a blackbody radiation curve is determined solely by the surface temperature of the blackbody.

(3) The colour of a star can be deduced from the peak of the blackbody radiation curve.

A (2) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

★**12** 3.3 Compared with the Sun, Star X gives out more radiation per unit surface area at every wavelength. However, the Sun is more luminous. Which of the following descriptions about star X is correct?

A Star X is larger than the Sun.
B Star X is a red giant star.
C Star X is bluer than the Sun.
D Star X has a lower surface temperature than the Sun.

(For Q13–14.) Apparent brightness is plotted against distance from the Earth for stars with luminosities L_1 and L_2 (Fig e). Consider stars P, Q and R on the curves.

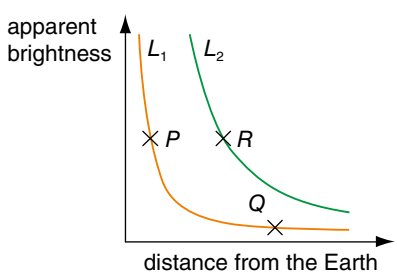


Fig e

★★**13** 3.2 Which star(s) has/have the largest value of apparent magnitude?

A Q B R
C P and R D P and Q

Refer p.107

★★**14** 3.2 Which star(s) has/have the smallest value of absolute magnitude?

A Q B R
C P and R D P and Q

Refer p.110

**15 ★★ 3.1 Jimmy took two photographs of star *P*, one in January and one in March. Figure f shows the positions of the Earth in these two months. Jimmy tries to use the photographs to determine the distance of *P* from the Earth.

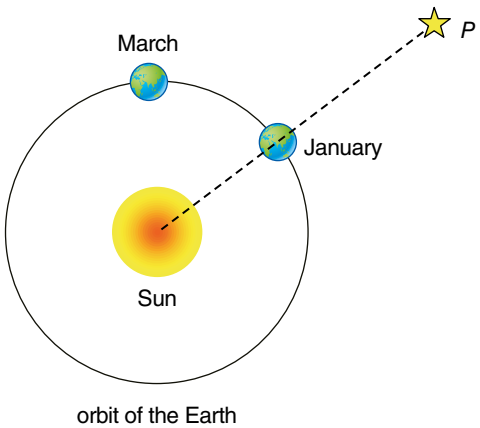


Fig f

He finds that the angular shift of star *P* in these two months is 0.15″. Which of the following statements is/are correct?

(1) If the photographs had been taken in April and October, the angular shift of star *P* would be larger.

(2) The distance between star *P* and the Earth is less than 13.3 pc.

(3) The parallax of star *P* is 0.15″.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

▶ Refer p.99–101

16 HKDSE 2016 Paper 2 Q1.7 Misc

Stars *X* and *Y* are of equal brightness to the naked eye. The measured parallax of star *X* is twice that of star *Y*. What is the ratio $\frac{\text{luminosity of star } X}{\text{luminosity of star } Y}$?

A $\frac{1}{4}$ (38%)

B $\frac{1}{2}$

C 2

D 4

17 HKDSE 2016 Paper 2 Q1.8 3.4

Three galaxies are separated by distance *D* as shown (Fig g). The H_α line of Galaxy *X* when observed from the Milky Way shows a red shift of $\Delta\lambda$.

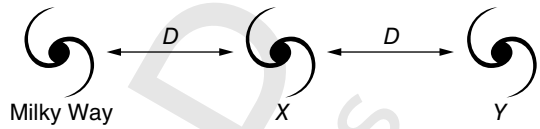


Fig g

Which of the following statement is/are correct?

(1) The H_α line of Galaxy *Y* when observed from the Milky Way shows a red shift greater than $\Delta\lambda$.

(2) The H_α line of the Milky Way when observed from Galaxy *X* shows no red shift.

(3) The speed at which Galaxy *X* is moving away from Galaxy *Y* equals the speed of which Galaxy *X* is moving away from the Milky Way.

A (2) only

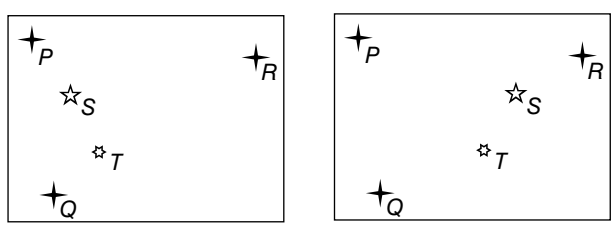
B (3) only

C (1) and (2) only

D (1) and (3) only (38%)

18 HKDSE 2017 Paper 2 Q1.4 3.1

Figure h shows two pictures of the same region of the sky taken in January and May of a certain year. *P*, *Q*, *R*, *S* and *T* are five stars.



January view *May view*

Fig h

Which of the following statements **must be** correct?

(1) Stars *P*, *Q* and *R* are equidistant from the Earth.

(2) The parallax of star *S* is smaller than that of star *T*.

(3) Star *S* is closer to the Earth than star *T*.

A (1) only

B (3) only (51%)

C (1) and (2) only

D (2) and (3) only

19 **HKDSE 2018 Paper 2 Q1.8**
Misc

Figure i shows the side view of a typical galaxy with a spherical nucleus of radius R and several spiral arms. Observations show that the stars in the nucleus are rotating about the galactic centre C as if the nucleus is a solid body. It is expected that the rotation of the stars in the spiral arms about C will obey Kepler's third law but instead the stars are observed to rotate with nearly the same constant speed.

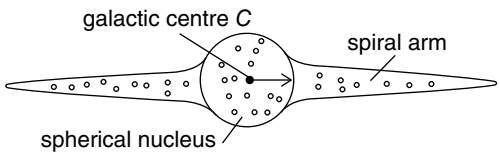
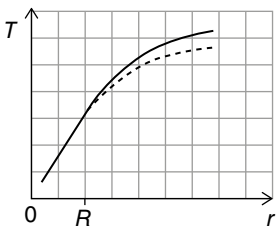
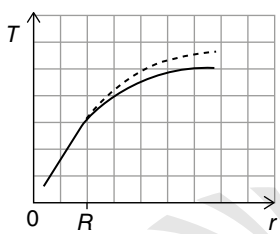
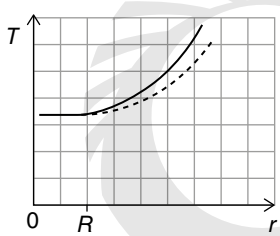


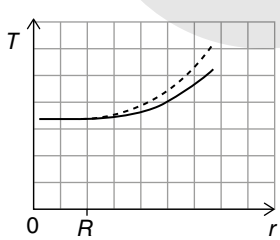
Fig i

Which graph best shows the variation of the rotational period T of the stars with their distance r from C? (The solid curve represents the observed result whereas the dotted curve represents the expected result.)

A


B


C


D (24%)


20 **HKDSE 2021 Paper 2 Q1.8**
Misc

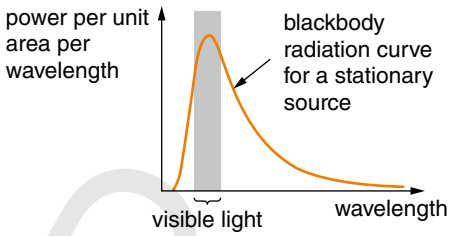


Fig j

Which of the following statements about the Doppler shift of the blackbody radiation from a source moving away from the Earth is/are correct?

(1) The peak of the blackbody radiation curve observed shifts to the right.

(2) The temperature of the source inferred from the observation is cooler than the actual value.

(3) The colour of the source observed looks different from that of a stationary source.

A (1) only
B (1) and (2) only
C (1) and (3) only
D (1), (2) and (3) (34%)

21 **HKDSE 2022 Paper 2 Q1.8**
Misc

The Hertzsprung-Russel (H-R) diagram below shows a series of main sequence stars.

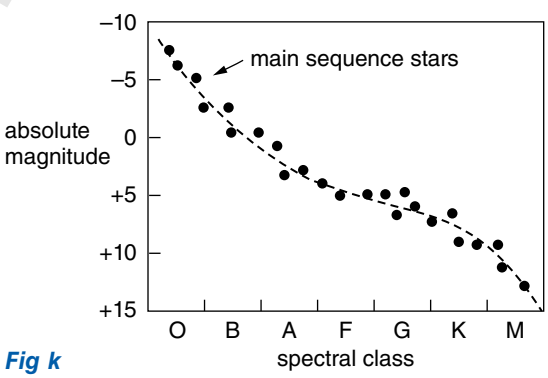


Fig k

The properties of three main sequence stars X, Y and Z are tabulated below (Table a). Which of them is/are further than 10 pc from the Earth?

Star	apparent magnitude	spectral class
X	0	F
Y	5	B
Z	4	K

Table a

A X only
B Y only (46%)
C X and Z only
D Y and Z only

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Conventional questions

22 3.2 The luminosity of the Sun is 3.84×10^{26} W. Its intensity measured on Earth is 1358 W m^{-2}.

(a) What is the luminosity of a star? (1 mark)

(b) What is the intensity of the Sun measured on Mars, which is 1.5 AU from the Sun? (2 marks)

23 Misc The wavelength of a spectral line from iron is 525 nm. In the spectrum of star *P*, the same spectral line appears as 525.17 nm. It is known that star *P* orbits a very massive star in a circular orbit.

(a) At what radial velocity is star *P* moving with respect to us? (2 marks)

(b) Would star *P* appear to be hotter or cooler than it actually is? Explain your answer. (2 marks)

24 Misc Stars *X* and *Y* have equal absolute magnitude. The intensity of *Y* is 10 000 times that of *X* as observed from the Earth.

(a) Calculate the ratio of the distance of star *X* to that of star *Y* from the Earth. (2 marks)

(b) Star *Y* has a stellar parallax of 0.08″. What is the stellar parallax of star *X*? (2 marks)

(c) Calculate the distances of stars *X* and *Y* from the Earth. (3 marks)

(d) The surface temperature of star *X* is 9 times that of *Y*. What is the ratio of their radius? Explain your answer. (3 marks)

(e) Star *X* is class B star. What type of star is *Y*? (1 mark)

***25** 3.2 Figure l shows a plot of apparent magnitude *m* against absolute magnitude *M*. The slope of the dotted line is 1. *Y* and *Z* are two stars.

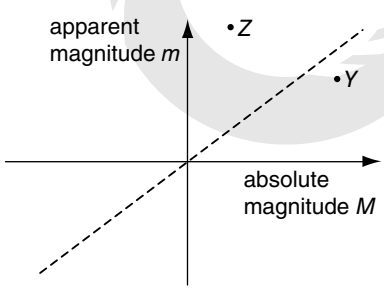


Fig l

(a) What are the distances to the stars that lie on the dotted line? Explain your answer. (2 marks)

(b) Explain whether the distance to *Z* is larger or smaller than the distance mentioned in (a). (2 marks)

(c) Hence, compare the distances to *Y* and *Z*. (1 mark)

***26** Misc Canopus is a giant star. It is the second brightest star in the night sky. The following data are obtained by direct measurements (Table b).

Intensity	5.24×10^{-8} W m^{-2}
Parallax *p*	0.0104″
Spectral type	F (surface temperature about 7400 K)

Table b

(a) Find the distance of Canopus from us. (2 marks)

(b) Find the luminosity of Canopus in terms of solar luminosity $L_\odot$, which is 3.84×10^{26} W. (2 marks)

(c) Find the radius of Canopus in terms of solar radius $R_\odot$. Take the solar surface temperature to be 5800 K. (2 marks)

***27** 3.4 Some dark lines are observed on the radiation spectra of a source on Earth and from a distant star (Fig m).



Fig m

(a) What kind of spectra are shown in Figure m? (1 mark)

(b) Explain briefly how the dark lines are formed. (2 marks)

(c) Explain why the position of the lines is different in the two spectra. (2 marks)

(d) What is the relationship between the radial velocity of the star and the positions of the dark lines? (2 marks)

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***28** 3.4 (a) Hubble's law can be mathematically presented as $v = Hd$, where H is a proportionality constant called Hubble constant. What do v and d represent? (2 marks)

(b) Figure n shows a graph of fractional shifts of wavelength $\left(\frac{\Delta\lambda}{\lambda_0}\right)$ of distant galaxies against d.

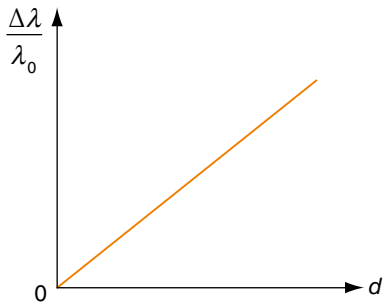


Fig n

(i) Express the slope of the graph in terms of H and c. (2 marks)

(ii) The spectrum of a galaxy contains a line with wavelength 659.6 nm. When measured in a laboratory, this line from a stationary source has a wavelength of 656.3 nm. Find the distance of the galaxy from the Earth in Mpc. Take $H = 73$ km s^{-1} Mpc^{-1}. (2 marks)

***29** 3.4 David observes the spectrum of a galaxy X. He then plots a graph of the observed wavelengths of galaxy X against that observed from a stationary source (Fig o).

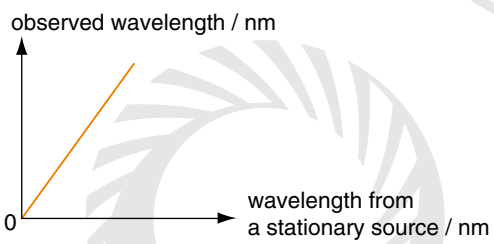


Fig o

(a) Explain the physical meaning of the slope when its value is (i) larger than 1, (ii) equal to 1 and (iii) smaller than 1. (3 marks)

(b) Suppose the value of the slope is 1.001. Calculate the radial velocity of galaxy X. (3 marks)

(c) State an assumption you made in (b). (1 mark)

****30** 3.1 Mercury is an inferior planet which has an elliptical orbit, whereas the orbit of the Earth can be considered circular (Fig p).

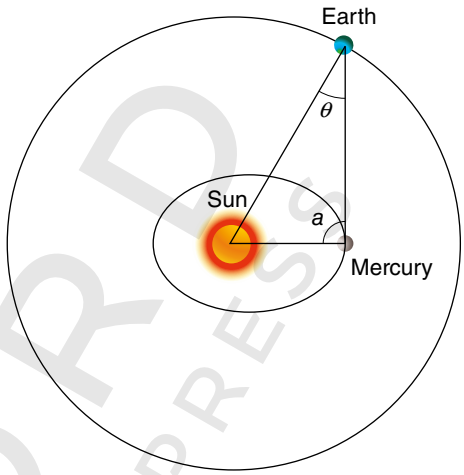


Fig p

Table c shows some date of Mercury.

Aphelion	0.47 AU
Perihelion	0.31 AU
Orbital period	88 days

Table c

(a) By using the sine formula, the following equation is obtained.

$$\frac{\sin a}{1} = \frac{\sin \theta}{0.47}$$

Find the maximum angular separation θ between Mercury and the Sun as observed from the Earth.

[Hint: sin a is maximum when $a = 90°$] (1 mark)

(b) If someone on Mercury wants to measure the distance between the Earth and Mercury. Is the parallax method suitable to use? Explain. (3 marks)

(c) The maximum angular size of Mercury as seen on Earth is 13″. Estimate the diameter (in km) of Mercury.

Given: 1 AU = 1.5×10^8 km

[Hint: For small angles measured in radians, tan $\theta \approx \theta$.] (3 marks)

Refer p.99–101

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**31 ★★ 3.4

Stars X and Y rotate about their common centre of mass C in uniform circular motion as shown (Fig q). Figure r shows the graph of radial velocity of X against time.

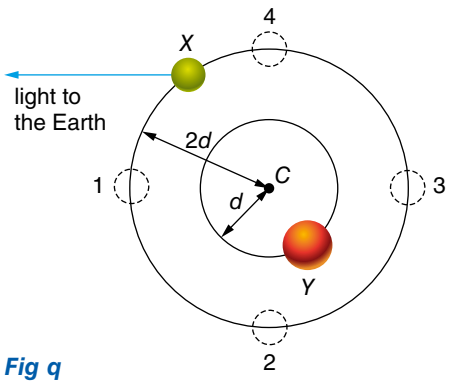


Fig q

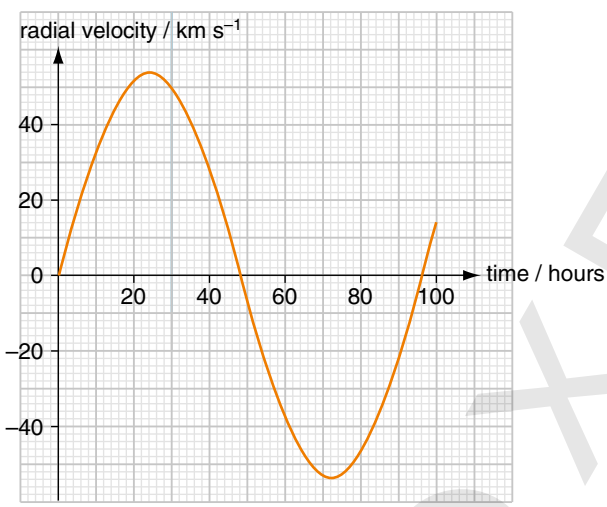


Fig r

(a) If X is at position 2 at time = 72 hours, what is the direction of orbital motion of the binary system? (1 mark)

(b) The standard wavelength of a spectral line is 656.28 nm. Find the wavelength of the same spectral line from X at time = 52 hours. (2 marks)

(c) (i) From Figure r, estimate the time for X to make one complete revolution around C. (1 mark)

(ii) Calculate the angular velocity and the orbital radius of X. (2 marks)

(d) (i) The radius of orbit of Y is half the radius of orbit of X. Sketch the graph of radial velocity of Y. (3 marks)

(ii) Find the mass of X (m_X) and the mass of Y (m_Y). (4 marks)

Refer p.131–135

**32 Misc

SQA 2017 AH Q6(b)-(c)

A Hertzsprung-Russel (H-R) diagram is shown in Figure s.

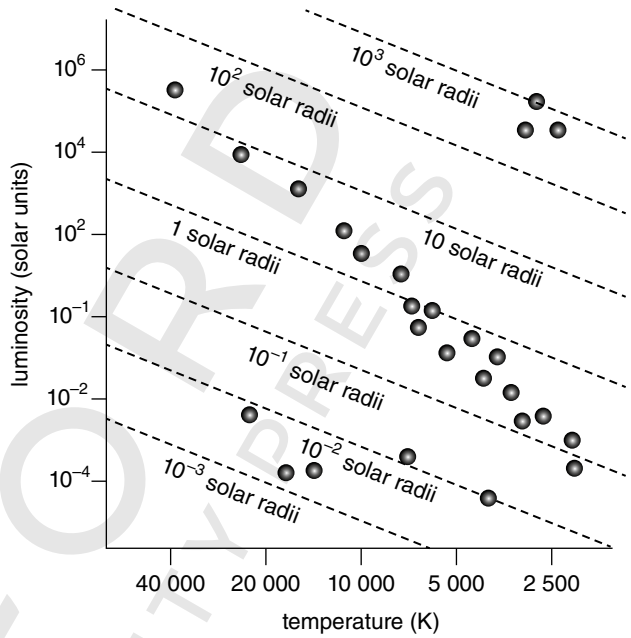


Fig s

(a) The luminosity of the Sun is 3.9×10^{26} W. The star Procyon B has a luminosity of 4.9×10^{-4} solar units and a radius of 1.2×10^{-2} solar radii.

(i) On the H-R diagram, circle the star at the position of Procyon B. (1 mark)

(ii) What type of star is Procyon B? (1 mark)

(iii) The apparent brightness of Procyon B when viewed from Earth is 1.3×10^{-12} W m^{-2}.

Calculate the distance of Procyon B from Earth. (4 marks)

(b) The expression

$$\frac{L}{L_0} = 1.5\left(\frac{M}{M_0}\right)^{3.5}$$

can be used to approximate the relationship between a star's mass M and its luminosity L.

L_0 is the luminosity of the Sun (1 solar unit) and M_0 is the mass of the Sun.

This expression is valid for stars of mass between $2M_0$ and $20M_0$.

Spica is a star which has mass $10.3M_0$.

Determine the approximate luminosity of Spica in solar units. (2 marks)

See p.156 for the Exam Report.

33 3.4 **HKDSE 2015 Paper 2 Q.1**

Figure t shows a distant binary star system viewed by an observer on Earth who is also on the orbital plane of the two stars. The system consists of stars 1 and 2 with masses m_1 and m_2 respectively orbiting in uniform circular motion about their centre of mass O under their mutual gravitational force. They move with **the same period** in two orbits of radii r_1 and r_2 with orbital speeds v_1 and v_2 respectively.

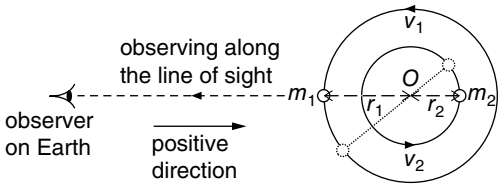


Fig t

By finding the radial velocity v_1 of the two stars inferred from the Doppler shift ($\Delta\lambda$) of the hydrogen-alpha line (H_α) observed on Earth, astronomers are able to deduce the masses of the stars. Assume that the centre of mass O of the binary system is stationary with respect to the observer. Figure u shows the radial velocity curves for the two stars. The direction moving away from the observer is taken to be positive velocity.

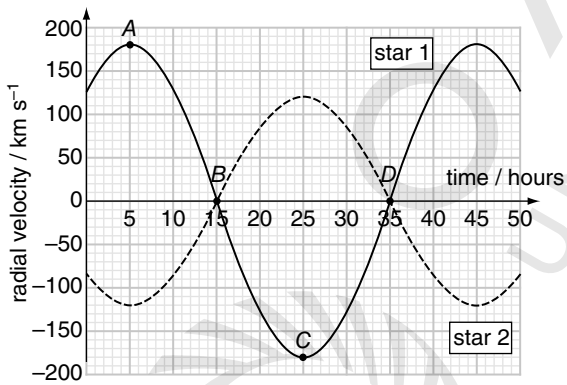


Fig u

(a) (i) What does it mean by **radial velocity** v_r of a star observed on Earth? (1 mark)

(ii) Identify which point, A, B, C or D, marked on the radial velocity curve corresponds to the orbital position of star 1 (in Solid line) **at the instant** shown in Figure t. (1 mark)

(b) Find, from Figure u, the orbital speed v_1 of star 1 and calculate its orbital radius r_1. Using a similar method, or otherwise, find the orbital radius r_2 of star 2. (4 marks)

(c) By considering the circular motion of star 1, calculate the mass m_2 of star 2. (2 marks)

(d) A spectrometer can only measure change of wavelength larger than 0.5 nm. Explain whether this instrument is suitable to measure the Doppler shift $\Delta\lambda$ of the hydrogen-alpha line (λ_0 = 656.28 nm) of the two stars. (2 marks)

See p.156 for the Exam Report.

34 Misc **HKDSE 2020 Paper 2 Q.1**

A certain star X at a distance of about 50 kpc from the Earth exploded a very long time ago and became Supernova 1987A (SN 1987A). The light from this supernova first reached the Earth in 1987.

(a) Estimate how long ago the above explosion took place. (Give your answer in years to 3 sig. fig.) (1 mark)

(b) At maximum brightness, SN 1987A has an apparent magnitude of +2.9. Is the absolute magnitude of SN 1987A at maximum brightness smaller than, larger than or equal to +2.9? Explain your answer. (2 marks)

Before the explosion of star X, its luminosity was about 40 000 times that of the Sun and its surface temperature was 3.1 times that of the Sun.

(c) (i) Use Stefan's law to show that the radius of the star X is about 20 times the Sun's radius. (2 marks)

(ii) In which region, A, B, C or D, on the Hertzsprung-Russell diagram was star X located? Explain whether or not this star can be classified as a 'red giant'. (2 marks)

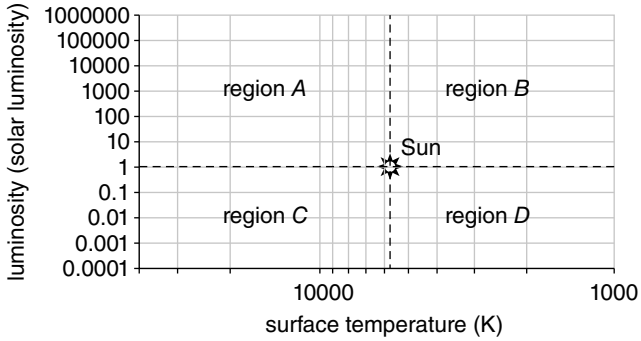


Fig v

(d) A special feature of SN 1987A is that a circular ring of gas surrounds the supernova. The gas was ejected by the star X some time before it exploded. As shown in Figure w, each point on this ring is expanding outwards at a constant speed from the supernova.

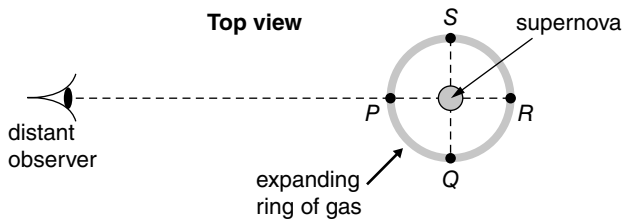


Fig w

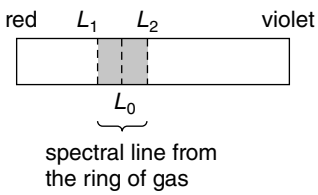


Fig x

Suppose a distant observer on the plane containing the ring views a certain spectral line from the ring of gas and finds that it covers wavelengths between the limits L_1 and L_2 as shown in Figure x. L_0 is the wavelength of that spectral line when observed in the laboratory. State the respective wavelengths that originate from point Q and point R on the ring. Explain your answer. (3 marks)

See p.156 for the Exam Report.

35 HKDSE 2021 Paper 2 Q.1

Misc

In our galaxy, there is a strong radio wave emitting source known as Sgr A* which is located at a distance 7940 pc away from the Earth. A star X is found orbiting around Sgr A* in an elliptical orbit with a period of 16.0 years.

(a) (i) The semi-major axis of the orbit, a, of star X is known to have an angular size of 0.125″. Determine the value of a in units of AU. (1 mark)

(ii) Hence use Kepler's third law for elliptical orbits, $T^2 = \dfrac{4\pi^2 a^3}{GM}$, to show that the mass of Sgr A* is about 3.82×10^6 times the mass of the Sun. (2 marks)

(b) As shown in Figure y, an observer on Earth is aligned with the semi-major axis of the elliptical orbit $ABCD$ of star X. The variation of the radial velocity v_r of X along the line of sight is shown in Figure z: v_r is taken to be positive for an object receding from the observer while a negative v_r implies an approaching object. The possible locations of Sgr A* are positions 1 or 2.

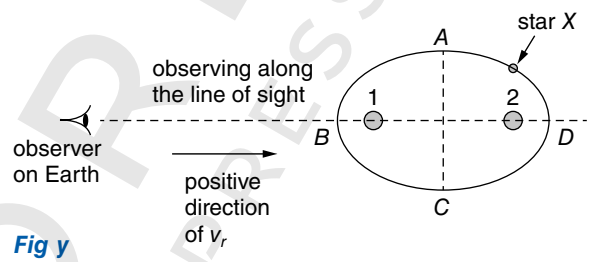


Fig y

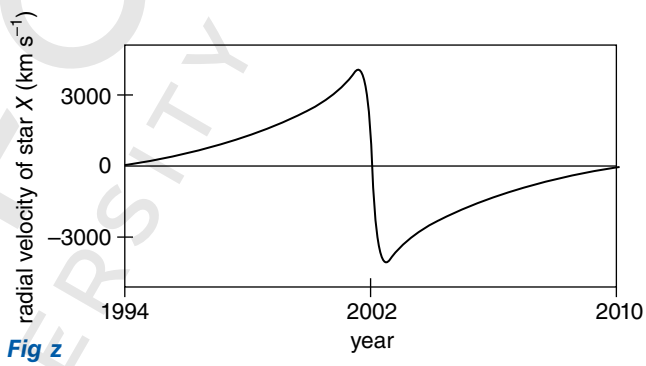


Fig z

(i) Give one method to determine v_r. State the difference in observation for positive and negative values of v_r in your proposed method. (2 marks)

(ii) State where star X is located, A, B, C or D, around the year 2002. Hence determine the location of Sgr A* (position 1 or position 2). Explain your choice. (2 marks)

(c) For a spherical celestial body of mass M and radius R, the escape velocity from its surface is given by $v = \sqrt{\dfrac{2GM}{R}}$, where G is the universal gravitational constant. Scientists believe that Sgr A* is a black hole, which is supposed to have an extremely strong gravitational field on its surface that even light cannot escape. Using the above equation and the result of (a)(ii), estimate the radius of this black hole (assume spherical mass distribution) in units of AU. Given $GM_S = 1.33 \times 10^{20}$ N m^2 kg^{-1}, where M_S is the mass of the Sun. (3 marks)

Know more

Other applications of parallax (p.99)

Parallax gives us perception of depth. Apart from measuring stellar distances, parallax can be applied in 3-D imaging and virtual reality (VR).

3-D imaging

To create the parallax effect, a special camera with two lenses separated by the same distance as our eyes is used. The lenses capture two images simultaneously, similar to those that each of our eyes perceives at the same time. When we look at a '3-D photograph' (or watch a '3-D movie'), we have to wear special glasses which ensure our eyes receive the respective images correctly.



Dual-lens 3-D camera recorder.



Special glasses for 3-D movie.

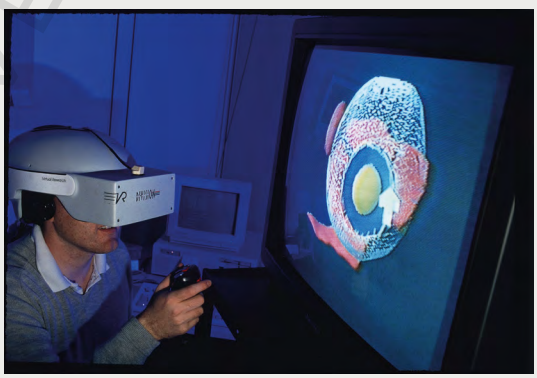
Virtual reality (VR)

For VR, we have to wear a headset with built-in screens. The two screens show two slightly different angles of the same image to each eye. Our brains interpret the image from each eye and give the perception of depth.

Nowadays, the use of VR has become increasingly popular. With sensors tracking our motions, we can interact with the 3-D simulated environment in a seemingly real way. This makes VR especially useful for educational learning and training.



Two screens in the VR headset.



Practicing virtual eye surgery.

The Harvard spectral classification (p.121)

The Harvard spectral classification was developed at Harvard College Observatory in the 1880s.

Initially, the astronomers in the observatory classified the stars into different spectral classes labelled alphabetically. *Annie Jump Cannon*, who also worked in the observatory, simplified the classification system and re-arranged the classes. In her lifetime, she classified more than 300 000 stars.



Annie Jump Cannon.

Annie Jump Cannon 坎農

Q33 Exam Report: Candidates' performance in (a) was fair. Most of them failed to state the concise meaning of radial velocity of a star observed on Earth. For the calculation of orbital radii in (b), candidates were able to use the information given in the graph. Many had difficulties in understanding the binary stars system and therefore failed to set up a correct equation of motion in (c). Some candidates even treated the circular motion of star 1 as if it was a single star. Most candidates managed to give correct explanations concerning the suitability of using the spectrometer in (d).

Q34 Exam Report: In (a), most candidates knew how to find when the explosion occurred although some of them made mistakes in converting units. In (b), some candidates failed to point out that absolute magnitude corresponds to observing the star as if it were placed at 10 pc. Part (c) was well answered. In (c)(ii), many knew the region on the H-R diagram in which star *X* was located. Most were able to identify features of star *X* that ruled out the possibility of classifying it as a 'red giant'. A few candidates forgot to start with Stefan's law. In (d), some candidates had a misconception that there was no Doppler effect as the distance of the star from the observer did not change.

Know more

No green or violet stars observed? (p.116)

Although the colour of stars is determined by the wavelength where the emission peaks, we seldom see green or violet stars. This can be explained by the perception of human eye.

For the stars that emit most radiation at the green part of visible spectrum, as green part is at the middle of the spectrum, roughly equal intensities of other colours of visible light are emitted as well. The stars are therefore perceived as white, which is a combination of all colours.

For the stars that emit most radiation at the violet part, high intensity of blue light is also emitted. As our eyes are more sensitive to blue light than violet light, these stars are perceived as blue.

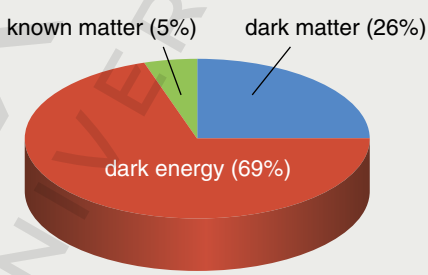


Constellation Orion.

Components of the universe (p.138)

We know that the universe contains planets, stars, nebulae, galaxies, gases, etc. However they only make up about 5 percent of mass-energy of the universe. The universe is more complex than we imagine.

Besides the known matter, dark matter and dark energy are proposed in the universe. Dark matter cannot be detected directly but indirect evidence supports the existence of dark matter (e.g. the rotation curve). Dark energy is the energy that makes the expansion of the universe accelerate. However, we do not know much about dark matter and dark energy at the moment.



Black hole

When a very massive star comes to the end of its life, it collapses under its own gravitational force so that a huge amount of mass contracts to form an extremely dense object. This object is black hole. The gravity of black hole is so strong that nothing, not even light, can escape from it.

The Doppler effect can also help in the search for black holes. Astronomers examine if there is any orbital motion of stars caused by a nearby unseen object. If the unseen object is massive and X-ray emission is detected, then the unseen object may be a black hole.



The first direct image of a black hole taken in 2019.

The X-ray emission is caused by the hot gas near the black hole. ▶

Q35 Exam Report: Part (a)(i) revealed that many candidates did not fully understand the concepts of parsec and angular size. Quite a number of them were not aware that the angle must be in radian when calculating the angular size. In (a)(ii), most candidates were able to start with Kepler's third law but less able ones failed to compare the Sgr A*-*X* and Sun-Earth systems in order to find the mass of Sgr A*. In (b)(i), candidates knew that the radial velocity could be deduced using the Doppler effect. In (b)(ii), most knew that star *X* was at *D* around the year 2002. However, few could correctly explain the fact that Sgr A* is at position 2. Some candidates confused radial velocity and orbital speed. In (c), a considerable number of them mistakenly used the radial velocity as the escape velocity to calculate the radius of the black hole.